Firm-Size Dispersion and the Limits of Age: Evidence from Swedish Administrative Data*

Markus Kondziella†

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Abstract

Using Swedish administrative data I document a sustained increase in firm-size dispersion over the past two decades. A decomposition based on the law of total variance shows that 89.2% of the increase is accounted for by rising size dispersion within firm-age groups, particularly among older firms, while changes in mean size by age and shifts in the age distribution play a limited role. In the cross-section, I find that within-age heterogeneity accounts for 99.4% of total variance: firm age predicts mean size but explains almost none of the variance. These findings discipline quantitative models of firm dynamics.

Keywords: Firm dynamics, Firm-size dispersion, Administrative data

JEL codes: D22, L11, L25, E23, O47

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†University of St. Gallen and Swiss Finance Institute. E-mail: markus.kondziella@unisg.ch

1 Introduction

Firm-size dispersion has risen substantially across advanced economies over the past decades (Autor et al., 2020; Akcigit and Ates, 2021).[1] Understanding what drives this trend matters: the forces behind rising dispersion point to very different underlying mechanisms depending on whether they operate through the aging of the firm population or the divergence of incumbent firms. This paper decomposes the rise in firm-size dispersion into three observable margins of firm dynamics and shows that virtually all cross-sectional size variation, and the overwhelming share of its rise, is accounted for by heterogeneity among firms of the same age.

I use comprehensive administrative data covering the universe of Swedish firms between 1997 and 2017. Applying the law of total variance, I decompose cross-sectional dispersion in firm size into three components: differences in mean size across firm ages, the firm age distribution, and size dispersion conditional on age. This decomposition is model-free and holds exactly in the data. Two findings stand out. First, the decomposition reveals that 89.2% of the rise in size dispersion since the turn of the millennium is driven by rising within-age dispersion, particularly among older incumbents, while the aging of the firm population contributes modestly (13.6%) and changes in mean size by age contribute negligibly (-2.8%). Second, decomposing the level of size dispersion in 2017, I find that 99.4% of cross-sectional variance is accounted for by within-age variation, with differences in mean size across ages accounting for the remaining 0.6%. Firm age strongly predicts the average size of a firm — firms aged 21+ are more than a log point larger than entrants — but explains almost none of its variance. The results hold across sectors and alternative size measures, including employment.

These findings sharpen and partly redirect the recent empirical literature on firm dynamics. Hopenhayn et al. (2022) and Karahan et al. (2024) document two facts for the US: the firm age distribution has shifted toward older firms, and mean size conditional on age has remained stable. I confirm both patterns in Swedish data. But these two margins together account for only about 11% of the rise in size dispersion. The dominant margin — within-age dispersion — has received comparatively little attention. Sweden's regulated labor market and strong welfare state might be expected to dampen size dispersion relative to less regulated economies, making the documented findings more striking rather than less.

The dominant role of within-age heterogeneity disciplines quantitative models of firm dynamics in two ways. For the cross-section, models calibrated to match mean size by age — the prevailing target — can substantially undershoot within-age dispersion. I illustrate this using the framework of Peters (2020): applying the same decomposition to the model's steady state yields a within-age share of 68.9%, more than 30 percentage points below the data. Targeting size dispersion conditional on age alongside the mean profile would provide

[1] See also Grullon et al. (2019); Decker et al. (2016*a*, 2020); Gourio et al. (2014); Chen et al. (2023). De Loecker et al. (2020) further document a reallocation of market shares from low- to high-markup firms, which raises the aggregate markup and is consistent with rising firm-size dispersion.

an additional disciplining moment. For the time series, the rise in within-age dispersion shifts attention from firm demographics to forces that further differentiate incumbents over time, such as productivity divergence (Aghion et al., 2023). I rule out two alternative drivers of rising within-age dispersion: compositional shifts at entry and weakening selection. Dispersion among entrants is stable throughout the sample, and the rise is weakest in the most capital-intensive sectors, inconsistent with zombie-firm survival under low interest rates.

Related Literature. Beyond the work cited above, this paper connects to Decker et al. (2016*b*), who document that dispersion in firm *growth rates* has declined. My finding of rising dispersion in firm size *levels* conditional on age is consistent with theirs if growth rates have become more persistent and type-driven — the same firms increasingly experience high (or low) growth. Decker et al. (2020) provide direct evidence that firm growth has become less responsive to shocks, supporting this interpretation.

This paper also relates to a growing literature that studies the drivers of recent macroeconomic trends, including rising firm-size dispersion, declining entry, and slowing productivity growth (Aghion et al., 2023; De Ridder, 2024; Akcigit and Ates, 2023; Liu et al., 2022; Peters and Walsh, 2021). Before asking which exogenous force is responsible, however, one must establish which margins of firm dynamics actually move. The decomposition in this paper provides exactly these prior disciplining moments: any proposed mechanism must generate rising within-age dispersion among incumbents as its prediction. Therefore, the decomposition directly constrains theory.

In Aghion et al. (2023), increasing productivity differences between incumbents incentivize more productive firms to expand into new product markets relative to less productive ones. Their model abstracts from firm entry and exit so mean size and dispersion conditional on age, as well as the age distribution are not defined.

The remainder of the paper is organized as follows. Section 2 introduces a model-free decomposition of firm-size dispersion based on the law of total variance. Section 3 describes the administrative data and measurement choices. Section 4 documents trends in firm-size dispersion and its underlying components. Section 5 quantifies the contribution of each margin using a decomposition exercise. Section 6 discusses the implications of the findings for quantitative models of firm dynamics. Section 7 concludes.

2 A decomposition of firm-size dispersion

To structure the empirical analysis, I begin with a model-free decomposition of firm-size dispersion. Consider the law of total variance, $\mathrm{Var}(Y) = \mathrm{Var}(\mathbb{E}[Y|X]) + \mathbb{E}[\mathrm{Var}(Y|X)]$, which decomposes the total dispersion in Y into (i) variation across groups defined by X and (ii) variation within those groups. Setting $Y \equiv \ln s_f$ and $X \equiv a_f$, where s_f and a_f denote firm size and age, respectively, this decomposition partitions cross-sectional dispersion in log firm size into two components: differences in average size across firm ages (the "between" component) and size dispersion conditional on age (the "within" component). Specifically,

$$\text{Var}\left(\ln s_f\right) = \underbrace{\sum_{a_f} p(a_f)\left[\mathbb{E}[\ln s_f|a_f] - \sum_{a_f} p(a_f)\mathbb{E}[\ln s_f|a_f]\right]^2}_{\text{Between}} + \underbrace{\sum_{a_f} p(a_f)\text{Var}(\ln s_f|a_f)}_{\text{Within}}, \tag{1}$$

where $p(a_f)$ denotes the share of firms of age a_f. The definition of firm age can be either discrete (e.g., integer years) or based on broader age bins; the decomposition applies in either case.

Although the decomposition uses the variance as the dispersion measure, it more generally reveals that cross-sectional firm-size dispersion is shaped by three margins: (i) differences in mean size across firm ages, $\mathbb{E}[\ln s_f|a_f]$; (ii) dispersion in size within age groups, $\text{Var}(\ln s_f|a_f)$; and (iii) the age distribution, $p(a_f)$. Accordingly, total dispersion can increase for three distinct reasons: greater divergence in mean size across ages, increased within-age heterogeneity, or shifts in the age distribution toward older firms, which tend to be larger and exhibit greater within-age dispersion.

How have these components evolved over time? The next sections examine each in turn. I first describe the data, then turn to the empirical trends.

3 Data

The data come from Statistics Sweden's main firm-level dataset, Företagens Ekonomi, which contains annual balance sheet and income statement information for the universe of Swedish firms over the period 1997–2017. It includes firm-level variables such as sales, assets, intermediate inputs, and employment, as well as information on industry and legal form. I restrict the sample to firms in the private sector that eventually employ at least one worker. For a detailed description of the data, see Section A in the Supplemental Appendix.

Firm age is defined as the number of years since the first worker — either employee or self-employed — joins the firm. This information comes from the auxiliary dataset Registerbaserad Arbetsmarknadsstatistik, which covers the universe of employer-employee matches and extends back to 1992. This allows me to observe untruncated firm age for all firms founded from 1993 onward.

All nominal variables are deflated to 2017 Swedish kronor (SEK) using the GDP deflator. The results are virtually unchanged when using sector-specific deflators; moreover, sectoral deflation does not affect measures of within-sector dispersion in log firm size.

The final dataset comprises 7,590,605 firm-year observations. Summary statistics are reported in Table A-1. The median firm has sales of approximately 1.4 million SEK (1 SEK ≈ 0.1 USD) and employs one full-time equivalent worker. The distributions of sales and inputs are highly right-skewed: mean sales (10.9 million SEK) and employment (4.9 workers) substantially exceed their respective 75th percentiles. A nontrivial fraction of firms report zero capital stock (fixed assets net of depreciation), employment, or wage bill, as reflected in zero-valued 25th percentiles for these variables. This feature has implications for the choice of the firm-size measure.

I use firm sales as the baseline measure of size and provide robustness checks using alternative proxies. Sales offer two advantages. First, the incidence of zero sales is negligible, avoiding the log-of-zero problem that arises with employment. Second, sales provide a more continuous measure of firm size, whereas employment is highly discrete. For example, the 25th, 50th, and 75th percentiles of employment are 0, 1, and 3, respectively, limiting the informativeness of dispersion measures such as the interquartile range (IQR). While the IQR is the primary dispersion measure used in the paper, I also report results using the standard deviation. To mitigate the influence of outliers, I winsorize sales at the 0.1 and 99.9 percent levels.[2]

4 Size dispersion and firm dynamics in the data

This section examines cross-sectional size dispersion and the three margins of the decomposition in eq. (1), in both levels and over time. All margins are documented in the same data, allowing for an exact decomposition of firm-size dispersion.

4.1 Firm-size dispersion

Figure 1 shows cross-sectional firm-size dispersion over time, measured by the interquartile range (IQR) of log sales. The IQR rises smoothly from 2.0 to 2.5 log points, implying that the 75th-to-25th percentile size ratio grew from 7.4 to 12.2.

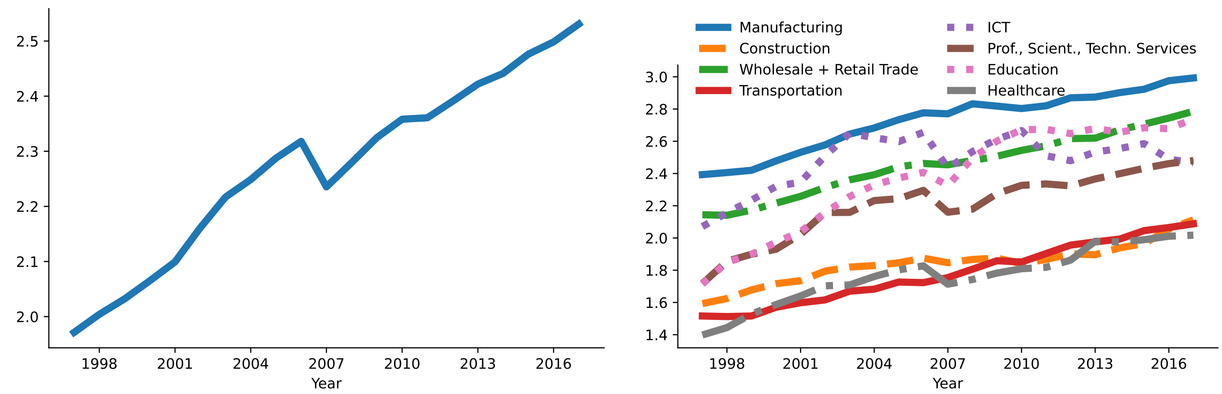
The right panel shows the same dispersion measure by sector. The increase in dispersion is shared across all sectors. The rate of increase is similar across sectors, with most displaying the same 0.5 log point rise shown in the left panel. Hence, the rise in aggregate dispersion is not driven by differential trends across sectors.

The advantage of the IQR is that it does not require dropping firms with zero sales, since the 25th percentile of the sales distribution is strictly positive. Other dispersion measures such as the standard deviation would require dropping observations with values of zero. Beyond this practical advantage, the IQR is robust to the extreme right-skewness of the sales distribution and captures dispersion in the bulk of the firm population rather than being driven by the tails. I show that results are unchanged when computing size dispersion using the standard deviation after dropping firms with zero sales. This is because changes in the tails are not systematically different from changes in the bulk of the distribution, and dropping firms with zero sales introduces negligible selection given its rarity. Figure 1 is virtually identical when using the standard deviation as the dispersion measure, as shown

[2] Throughout, sales are interpreted as a proxy for firm size rather than market share.

Figure 1: Cross-sectional firm-size dispersion

(a) All firms (b) By sector



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution. The left panel shows the IQR for all firms; the right panel shows the IQR by sector.

in Figure A-1. The standard deviation increases by 0.4 log points, close to the 0.5 log point increase in the IQR.

4.2 Firm dynamics

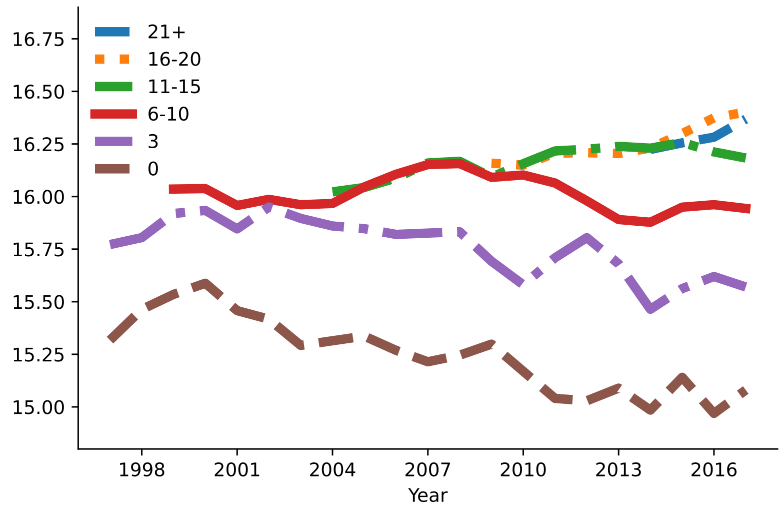
How have the three margins in the decomposition in eq. (1) — average firm size conditional on age, the age distribution, or size dispersion conditional on age — changed over time? The following subsections examine each margin in turn.

4.2.1 Firm size conditional on age

Figure 2 shows the log of the average firm size by firm age.[3] Two patterns are noteworthy. First, there are stark level differences across ages: older firms are, on average, much larger than younger firms. Firms aged 6–10 are on average about 0.75 log points larger than entrants. These differences vanish for very old firms, as firms aged 21 and older are not systematically larger than those aged 11–15. Second, mean size conditional on age is stable over time. Young firms, represented in the graph by firms aged zero or three, show a relatively stable or even slightly negative trend in average size. Firms of older ages show no systematic trend over time. Similar patterns were first documented by Hopenhayn et al. (2022) and Karahan et al. (2024) for the US and by Engbom (2023) for Sweden.

[3]Note that the decomposition in eq. (1) contains the average of the log firm size rather than the log of the average firm size. I show the latter to avoid dropping firms with zero sales.

Figure 2: Firm size conditional on age



Notes: The figure shows the log of the average firm size (sales) by firm age indicated in the legend. Sales are in 2017 SEK.

4.2.2 Firm age distribution

Figure 3 shows two margins of the firm age distribution: the startup rate in the left panel and the share of mature firms in the right panel. The startup rate, defined as the share of firms at most one year old, declined from 14.9% in 1997 to 10.4% in 2017. Hence, the share of young firms declined. At the same time, the share of mature firms, defined as firms at least ten years old, increased. I report the series from 2002 onward, as firm age (10+ years) is untruncated only from that year. The share of mature firms increased from 44.9% in 2002 to 56.4% in 2017, an increase of 11.5 percentage points over 15 years. Hopenhayn et al. (2022) and Karahan et al. (2024) previously documented a quantitatively similar shift in the firm age distribution for the US.

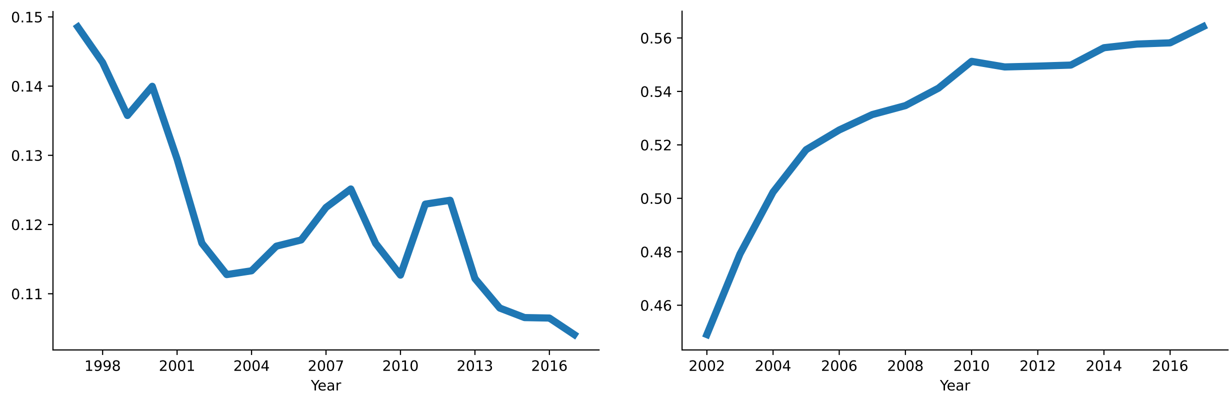
4.2.3 Firm size dispersion conditional on age

The left panel of Figure 4 shows the third and last element of the dispersion decomposition in eq. (1), namely size dispersion conditional on age, again measured using the IQR of log sales. Two points are noteworthy. First, there are stark level differences across ages: among older firms, sizes are much more dispersed than among younger firms. Second, dispersion has risen systematically over time, but the increase is concentrated among older firms. For ages 0 and 3, size dispersion was stable throughout the sample period. For age groups 6–10, 11–15, and 16–20, within-group size dispersion increased by 0.3 to 0.4 log points.

The right panel of Figure 4 shows size dispersion within the age group 11–15 by sector, with all sectors normalized to zero in 1997. All sectors show an increase in size dispersion conditional on age. The increase is quantitatively similar across sectors, with most displaying a rise of 0.3 to 0.4 log points consistent with the rise shown in the left panel for this age group. Among the sectors, Manufacturing displays the smallest increase whereas Healthcare shows the largest. Hence, the rise in firm-size dispersion conditional on age is a within-sector phenomenon. The pervasiveness of the trend across service sectors — whose output is largely non-tradable — further rules out foreign demand as a driver.

Figure 3: Firm age distribution

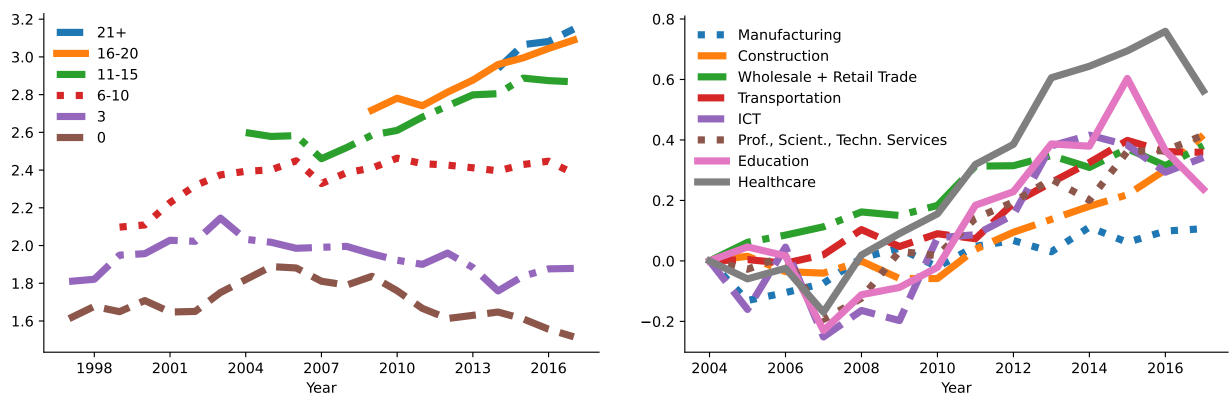
(a) Startup rate (b) Share mature firms



Notes: The left panel shows the startup rate defined as the share of firms at most one year old. The right panel shows the share of firms that are at least ten years old.

Figure 4: Firm size dispersion conditional on age

(a) All firms (b) By sector (ages 11-15)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution. The left panel shows the IQR for all firms conditional on age as indicated in the legend; the right panel shows the IQR by sector for firms aged 11-15.

These trends are robust to a range of specification choices, as shown in Supplemental Appendix B, including using the standard deviation of log sales conditional on age as the dispersion measure (excluding firms with zero sales), using integer ages as the conditioning variable without any age groupings, using sector-level GDP deflators (instead of the aggregate GDP deflator) to deflate firm sales, excluding mergers and acquisitions, and using alternative measures of firm size such as intermediate inputs, capital, or employment.

5 Decomposing firm-size dispersion

The graphical analysis in the previous section documented a shift in the firm age distribution toward older firms and a rise in size dispersion conditional on age. How much does each margin contribute quantitatively to the rise in cross-sectional size dispersion?

In this section, I decompose the rise in firm-size dispersion into its components. To this end, I fix two points in time, compute the standard deviation of log firm size using eq. (1) and then separately change either the average size conditional on age, the firm age distribution, or size dispersion conditional on age. I do so twice; once starting from the initial point and changing each component to its values in the end period and vice versa. I report averages across both. Because the decomposition is nonlinear, the contributions of the three margins do not necessarily sum to the total change, but in practice they do up to the fourth decimal. For this decomposition, I restrict to firms with positive sales so that I can implement the decomposition in eq. (1) exactly. As shown before, this is not consequential when using firm sales as the size measure.

The end point is naturally the last year in the data, 2017. The start point is less obvious. The further back in time, the longer the period one can decompose; however, firm age is untruncated only for firms born 1993 or later, so an earlier starting date requires a coarser age grouping for older firms. I choose 2002 as the starting point, which allows me to decompose a period of 15 years, from 2002 to 2017, and to differentiate firm ages up to age ten, i.e., in the decomposition in eq. (1), $a_f \in \{0, 1, \ldots, 8, 9, 10+\}$. Note that the decomposition in eq. (1) holds for any age grouping.

Between 2002 and 2017, dispersion of firm sales increased by 0.202 log points, as shown for the standard deviation in Figure A-1. Changes in mean size conditional on age contributed negatively to the total change by -0.006 log points (or -2.8%). This is consistent with average size conditional on age declining slightly in Figure 2. The shift in the age distribution contributed positively but modestly, at 0.027 log points (13.6% of the total). Lastly, the increase in size dispersion conditional on age contributed 0.18 log points, accounting for the largest part (89.2%). Rising within-age heterogeneity thus accounts for the overwhelming majority of the increase in size dispersion. Panel A of Table 1 shows that this result is robust across sectors. In most sectors, changes in mean size conditional on age make a small negative contribution, while changes in the age distribution contribute modestly and positively. By contrast, changes in size dispersion conditional on age are large and positive.[4]

[4]The ICT sector, where the largest part is accounted for by the aging of firms, is the only exception. This is specific to the decomposed time period: the starting year of the decomposition (2002) falls into the

Table 1: Decomposition of firm-size dispersion

Panel A: Decomposition of the change in $\sqrt{\mathrm{Var}(\ln s_f)}$ *(2002–2017)*				
	$\mathbb{E}[\ln s_f \mid a_f]$	$p(a_f)$	$\mathrm{Var}(\ln s_f \mid a_f)$	Total
Total economy	-0.006	0.027	0.180	0.202
Manufacturing	-0.016	0.052	0.226	0.262
Construction	-0.004	-0.006	0.275	0.265
Wholesale + Retail Trade	-0.009	0.031	0.271	0.293
Transportation	-0.007	0.028	0.276	0.297
ICT	-0.008	0.096	0.060	0.149
Prof., Scient., Techn. Services	0.008	0.023	0.158	0.189
Education	0.001	0.050	0.215	0.266
Healthcare	0.009	-0.008	0.187	0.187

Panel B: Level decomposition of $\mathrm{Var}(\ln s_f)$ *(2017)*	
	Share within (%)
Total economy	99.4
Manufacturing	98.3
Construction	99.5
Wholesale + Retail Trade	99.2
Transportation	98.9
ICT	99.4
Prof., Scient., Techn. Services	98.5
Education	98.2
Healthcare	98.0

Notes: Panel A decomposes the change in the standard deviation of log firm sales, $\sqrt{\mathrm{Var}(\ln s_f)}$, between 2002 and 2017. Each component is computed by varying one element at a time—the conditional mean $\mathbb{E}[\ln s_f \mid a_f]$, the age distribution $p(a_f)$, and within-age dispersion $\mathrm{Var}(\ln s_f \mid a_f)$—while holding the others fixed. Each element is varied relative to the start and end period and the table reports the average of both. Panel B decomposes the variance of log firm sales, $\mathrm{Var}(\ln s_f)$, in 2017 into the within and between component according to eq. (1) and shows the share of the within component.

The dominant role of within-age dispersion for the rise in firm-size dispersion raises an analogous question for its level: what share of total cross-sectional variance in a given year is accounted for by within-age heterogeneity? A complementary exercise decomposes the *level* of firm-size dispersion into the between and within components of eq. (1), giving a quantitative account of how cross-sectional size dispersion arises. I choose the end year of my data, 2017, for this decomposition, as the age distribution is least truncated then. I partition firm ages into integers up to age 24, after which firms are grouped, i.e., $a_f \in \{0, 1, \dots, 24, 25+\}$. I find that 99.4% of the variance of log sales $\text{Var}(\ln s_f)$ is accounted for by the within component. The decomposition of the variance is linear in the between and within components, so the between component accounts for the residual 0.6%. Hence, the overwhelming share of size dispersion reflects size dispersion conditional on age rather than size differences across ages. Panel B of Table 1 further shows that the share of the within component is 98% or higher in all sectors.

This pattern is not specific to sales as the size measure. Repeating the level decomposition using employment (restricting to firms with positive employment) yields a within share of 95.7%, indicating that the high within share reflects genuine size heterogeneity among same-aged firms rather than dispersion in markups that would only show up in sales.

The split between the between and within components depends on the age grouping used: the coarser the age grouping, the larger the within component. In the current decomposition, firm ages take integer values up to age 25. Any coarser age grouping would increase the within share even further.[5]

The result that the within component drives the level of dispersion is not obvious ex ante given that the firm size-age gradient is substantial. Figure 2 showed that firms above the age of 20 are — on average — more than a log point larger than entrants in 2017. Thus, age is highly informative about a firm's expected size, yet nearly uninformative about where in the size distribution it actually falls.

6 Discussion

The dominant role of within-age heterogeneity — both in the level and in the rise of firm-size dispersion — disciplines quantitative models of firm dynamics. This section evaluates two implications.

Standard calibrations miss within-age dispersion. Quantitative firm-dynamics models are typically calibrated to match the mean size-by-age profile. To assess how much

dotcom boom, which saw an increase in entrant activity in the ICT sector. The ICT sector was therefore characterized by a relatively young firm population at that time, which experienced substantial aging over the subsequent period.

[5]The age grouping does not systematically favor any component when decomposing *changes* in total firm-size dispersion. For the changes decomposition, what matters instead is how each component evolves over time.

within-age dispersion such calibrations generate, I solve the steady state of Peters (2020) using the paper's calibration and apply the decomposition in eq. (1).[6] Peters (2020) is a useful benchmark: the calibration explicitly targets the mean size-by-age profile, and the model incorporates state-of-the-art features such as endogenous markups. The decomposition yields a within-age share of 68.9%, more than 30 percentage points below the empirical 99.4%. The model assigns too large a role to life-cycle growth relative to within-age heterogeneity. Matching mean size by age is therefore insufficient: within-age dispersion is quantitatively the dominant moment and should be targeted alongside the mean profile.[7]

What drives rising within-age dispersion? The age profile of the rise is the key discriminating fact: size dispersion conditional on age increased substantially among older firms while remaining stable among young firms throughout the sample period (Figure 4). This pattern points to forces that accumulate with firm tenure. Two alternative explanations are inconsistent with the data.

First, weakening selection among unproductive firms — for instance, due to falling interest rates keeping otherwise non-viable firms alive — predicts that the effect should be strongest in capital-intensive sectors reliant on debt financing, and that it should primarily inflate the left tail of the size distribution by keeping small firms alive. Both predictions fail. The rise in size dispersion conditional on age is weakest precisely in Manufacturing (Figure 4), the most capital-intensive sector in the data. Moreover, as shown separately for each tail in Figure A-7, the 75th percentile of log sales rises while the 25th percentile falls. The simultaneous expansion of both tails is more consistent with genuine productivity divergence than with zombie-firm survival.

Second, rising size dispersion conditional on age could reflect changes in the composition of entry cohorts over time rather than post-entry divergence among incumbents. Figure 4 rules this out: size dispersion among young firms remained stable relative to dispersion among older firms throughout the sample period, implying that divergence accumulates after entry rather than being inherited from entry conditions.

As an illustration that rising productivity divergence is consistent with the documented trends, I introduce permanent productivity heterogeneity into the Peters (2020) framework (Supplemental Appendix C).[8] A rise in the productivity gap between high- and low-type firms generates an increase in size dispersion concentrated among older firms while leaving young-firm dispersion nearly flat, mirroring the empirical age profile; Supplemental Appendix C shows that even a modest rise in the productivity gap can quantitatively reproduce the 0.2 log-point increase among firms aged 21 and older in Figure A-2. In the model, rising

[6] I show how the decomposition maps to (an extended version of) the model in Appendix C.6.4.

[7] Future research could repeat this exercise for other models such as Hopenhayn (1992) or Luttmer (2007). In the Jovanovic (1982) passive-learning model, firms gradually discover their type after entry, raising the importance of the between component further.

[8] Alternative forms of ex-ante heterogeneity, such as differential intangible capital adoption (De Ridder, 2024), would work similarly.

productivity differences imply that growth rates become more type-driven and less determined by luck, which rationalizes rising dispersion in size levels conditional on age even if dispersion in growth rates falls (Decker et al., 2016*b*).

7 Conclusion

This paper documents a sustained rise in firm-size dispersion in Sweden from 1997 to 2017 and decomposes it into three margins of firm dynamics: differences in mean size across firm ages, the firm age distribution, and size dispersion conditional on age. I show that rising within-age heterogeneity accounts for 89.2% of the increase in dispersion between 2002 and 2017, while the aging of the firm population contributes modestly and changes in mean size by age contribute negligibly. Decomposing the *level* of size dispersion in 2017 reveals that 99.4% of total variance is attributable to within-age variation, implying that firm age explains almost none of the observed size differences in the cross-section.

The dominant role of within-age heterogeneity carries two broad implications. For the cross-section, the calibration exercise in Section 6 illustrates that targeting mean size by age alone is insufficient to account for the high within-age share documented here. Targeting size dispersion conditional on age directly alongside the mean profile would better discipline the mechanisms generating heterogeneity among same-aged firms. For the time series, theories of rising firm-size dispersion should center on forces that amplify heterogeneity among incumbents of similar age — such as rising productivity dispersion — rather than on declining entry or demographic aging alone.

The dispersion decomposition in this paper is model-free and holds exactly in the data, providing disciplining moments for both the cross-section and the time series for future quantitative work on firm dynamics. A natural complement would be to apply the decomposition to other countries and earlier periods, mapping how the three margins of firm dynamics have evolved across time and space.

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Supplemental Appendix for

"Firm-Size Dispersion and the Limits of Age: Evidence from Swedish Administrative Data"

[FOR ONLINE PUBLICATION ONLY]

A Data

The main dataset, *Företagens Ekonomi* (FEK), contains information from balance sheets and profit and loss statements for the universe of Swedish firms. From this dataset, I obtain the firm-size measures including sales, intermediate inputs, the capital stock, employment and the wage bill. In the FEK code-book by Statistics Sweden (SCB), these variables are defined as follows.[9] SCB defines sales as income from the companies' main business for goods sold and provided services. As the measure of the firm's capital stock I use fixed assets minus depreciation. Employment refers to the average number of employees in full-time units in accordance with the company's annual report. The measure of the firm's wage bill includes salaries and other remuneration, including severance pay. As described in the main text, I focus on firms in the private sector. These firms have a legal type (variable name: *JurForm*) less than 50 or equal to 96. Table A-1 contains summary statistics. The statistics are described in the main text.

Starting in 2004, the data include unincorporated self-employed with negative profits. These firms were previously not recorded, leading to an inflow of new firms in that year. Since I cannot differentiate between firms established in 2004 and firms entering the data in 2004 with an earlier birth year, I drop firms that enter the data in 2004 with negative profits when showing the firm age distribution. When showing the size-by-age patterns, I simply leave the 2004 cohort out.[10] For the main exercise — which decomposes the change in size

[9] `https://www.scb.se/contentassets/9dd20ce462644cc19f6f04eb2edbbe28/nv0109_kd_slut2022_v1_20240510.pdf`, accessed 26.03.2025.

[10] Leaving the 2004 cohort out when computing the age distribution would bias it in all years following

Table A-1: Summary statistics

	25th Pct.	50th Pct.	75th Pct.	Mean	SD
*Sales**	0.4	1.4	4.3	10.9	62.1
*Intermediate Inputs**	0.2	0.6	1.8	4.9	34.8
*Capital stock**	0.0	0.2	0.8	4.6	96.2
Employment (full-time units)	0	1	3	4.9	29.3
*Wage bill**	0.0	0.2	0.9	1.8	11.0

Notes: variables marked with * are in units of million 2017-SEK (1 SEK ≈ 0.1 US dollars). The capital stock is defined as fixed assets minus depreciation. The data include 7,590,605 firm-year observations.

dispersion between 2002 and 2017 — this is inconsequential: firm ages are grouped, so firms born in 2004 fall into the highest age group in 2017 regardless.

To measure size dispersion within sectors, I use the industry classification (SNI codes) provided by SCB. The industry classification changed twice between 1997 and 2017, once in 2002 and once in 2007. The changes in industry classification mainly affect the more detailed codes at the five-digit level, whereas I use the ones at the one-digit level for the main analysis. Nevertheless, to ensure a consistent industry classification over time I proceed as follows. During the year of the change, I observe both the old and the new industry classifications. For the firms present in the data in the year of the classification change, extending the new industry classification further back in time before the change is straightforward. This way, the industry codes of almost all firms are updated. A firm might be in the data before and after the classification change but not for the year of the change. For these firms, the above method does not work. If the firm appears in the data one year after the classification change, I use the observed classification after the change to update the classification before the change. For firms that are absent for several years around the year of change, I use industry mappings provided by Statistics Sweden. These mappings do not always provide a 1:1 mapping between industries before and after the classification change, so I use the most common transitions for the many-to-many mappings.

One concern is that changes in the firm structure, e.g., when firms merge with other firms, change the firm ID. To address this concern, I impute changes in firm IDs using worker flows between firms. The auxiliary data set *Registerbaserad Arbetsmarknadsstatistik* (RAMS) contains the universe of employer-employee matches. I impute changes in the firm ID of firms with at least five employees as follows: if more than 50% of the workforce of firm A in year t makes up more than 50% of the workforce of firm B in year $t+1$, I replace firm B's firm ID with firm A's firm ID from $t+1$ onward. The empirical results remain unchanged when

2004.

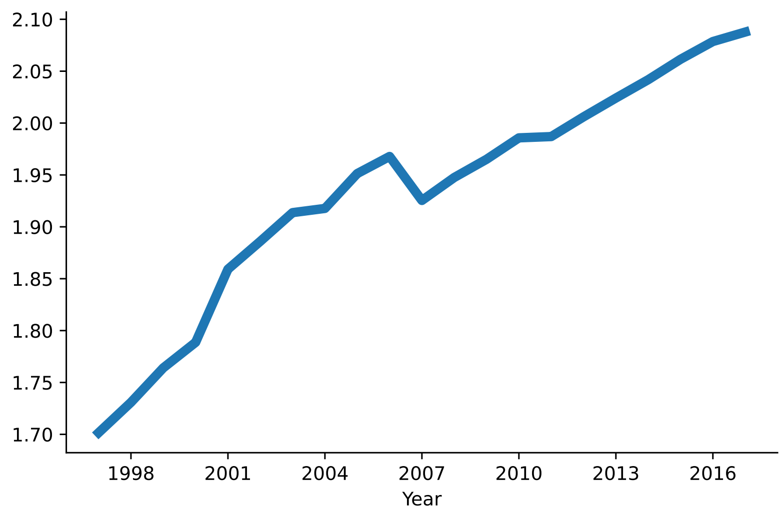
excluding firms for whom the imputed firm ID differs from the observed firm ID.

B Size dispersion and firm dynamics in the data

B.1 Firm-size dispersion

Figure A-1 restricts to firms with positive sales and plots the standard deviation of log sales. The standard deviation increases by about 0.4 log points.

Figure A-1: Firm size dispersion (standard deviation)



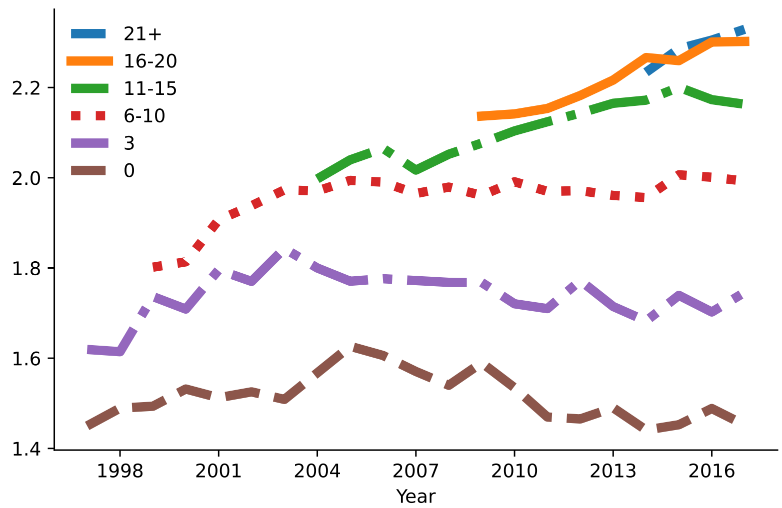
Notes: The figure shows the standard deviation of log sales, restricting to firm-year observations with positive sales.

B.2 Firm-size dispersion conditional on age

As a first robustness check, I use the standard deviation instead of the IQR. This is unproblematic for sales since zero sales are very infrequent, so restricting to positive observations is not consequential. For other size measures such as employment or the wage bill, restricting to positive values is more consequential and introduces stronger selection — self-employed workers, for instance, commonly report a zero wage bill.

As shown in Figure A-2, restricting to firms with positive sales and using the standard deviation of log sales as the measure of dispersion leaves the trends unchanged. Size dispersion conditional on age remains relatively stable for young firms and increases by about 0.2 log points for the age groups 6-10, 11-15 and 16-20.

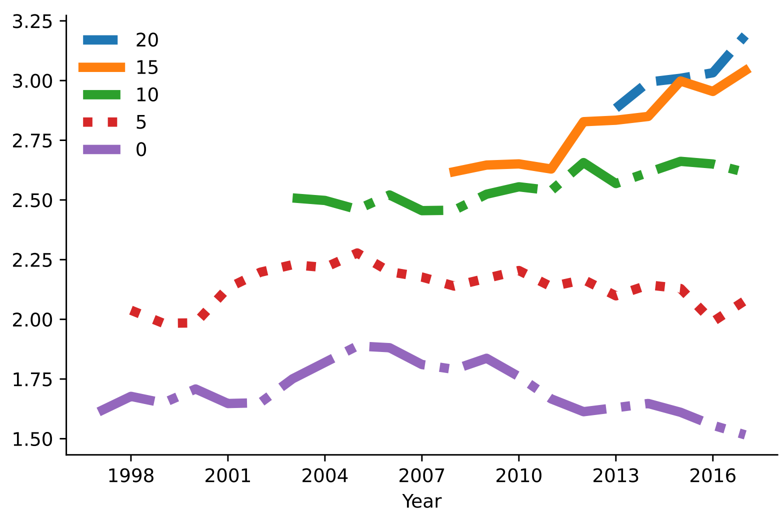
Figure A-2: Size dispersion conditional on age (standard deviation)



Notes: The figure shows the standard deviation of log sales by firm age as indicated in the legend. The figure restricts to firm-year observations with positive sales.

I next show that the age grouping is inconsequential. Figure A-3 shows size dispersion conditional on age for firms aged zero, five, ten, fifteen and twenty. The patterns remain the same. Size dispersion remained stable among young firms, but increased systematically among old firms.

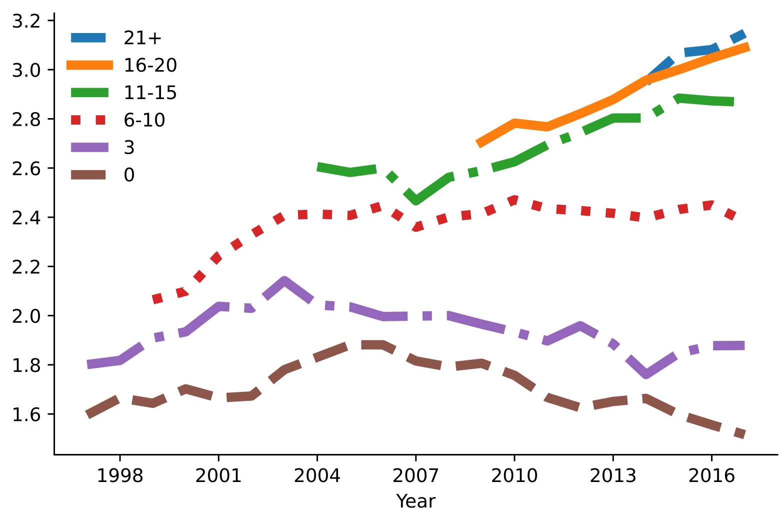
Figure A-3: Size dispersion conditional on age (alternative ages)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend.

The picture also remains unaffected when using alternative (finer) price deflators to deflate firm sales. Figure A-4 uses sector-level GDP deflators to deflate firm sales to 2017 SEK values. Note that these finer deflators have no effect on size dispersion within sectors, but could potentially have an effect at the aggregate level. Figure A-4 shows that this is not the case. The figure is virtually unchanged relative to the one in the main text. Differential inflation trends across sectors have no effect on changes in log sales dispersion at the aggregate level.

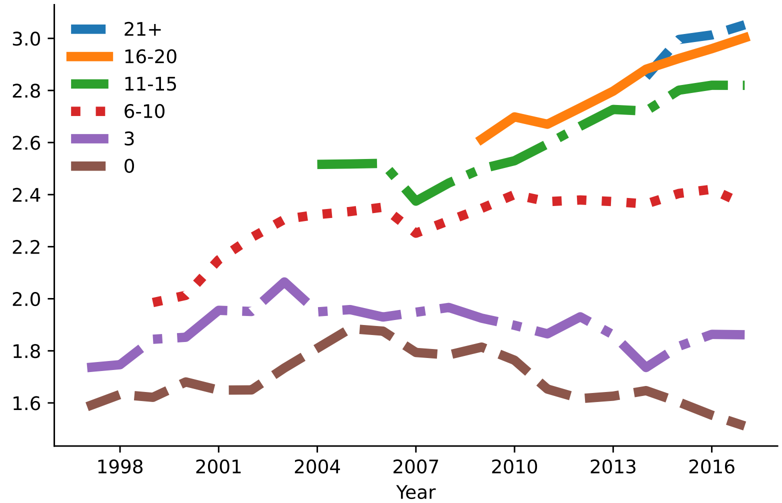
Figure A-4: Size dispersion conditional on age (sector deflators)



Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend. Sales are deflated using sector-specific GDP deflators.

I also show that mergers and acquisitions have not contributed to the rise in firm-size dispersion conditional on age. In the data, any change in firm structure renders a new firm ID. I identify mergers between firms (and any other changes in firm structure) based on worker flows as described in Appendix A and connect the firm IDs for identified cases. Figure A-5 shows size dispersion conditional on age when excluding firms for which the observed firm ID differs from the imputed one. The graph remains virtually unchanged.

Figure A-5: Size dispersion conditional on age (excluding mergers)



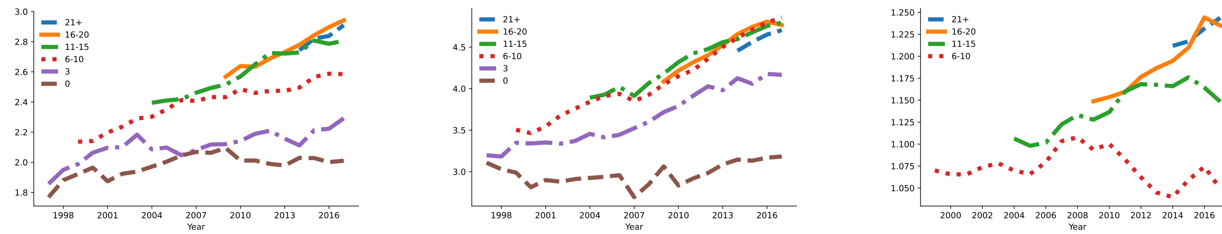
Notes: The figure shows the interquartile range (IQR) defined as the difference between the 75th and 25th percentile of the log sales distribution by firm age as indicated in the legend. The figure excludes firms with an imputed firm ID change.

Next, I assess the robustness of the documented increase in firm-size dispersion conditional on age using alternative size measures. As explained in the main text, the advantage of using firm sales as the main size measure is that sales are a continuous measure of size and the share of firms reporting zero sales is very low, so restricting to positive values is not consequential.

I document trends in size dispersion conditional on age for three alternative size measures: intermediate inputs, capital stock, and employment. As intermediate inputs are a firm's most frequently used input, the share of firms reporting zero intermediate inputs is also relatively low, so the standard dispersion measures can be applied without problems. The capital stock and employment pose more problems. When using the capital stock as the firm-size measure, the 25th percentile of the size distribution commonly features a capital stock of zero, making the IQR of log capital undefined. To circumvent this problem, I instead show the difference in the log capital stock between the 95th and 50th percentile. Hence, this dispersion measure focuses on dispersion in the right tail. For employment, the same problem occurs. The 25th percentile of the employment distribution generally features zero employment as self-employed do not hire any workers. An additional complication is that employment is a very discrete measure of firm size for the majority of firms (recall that the median firm employs one full-time worker). The IQR therefore does not accurately reflect changes in size dispersion. To make progress, when using employment as the size measure I exclude firms with zero employment and use the standard deviation as the dispersion measure.

Figure A-6: Size dispersion conditional on age (alternative size measures)

(a) Intermediate inputs (b) Capital stock (c) Employment

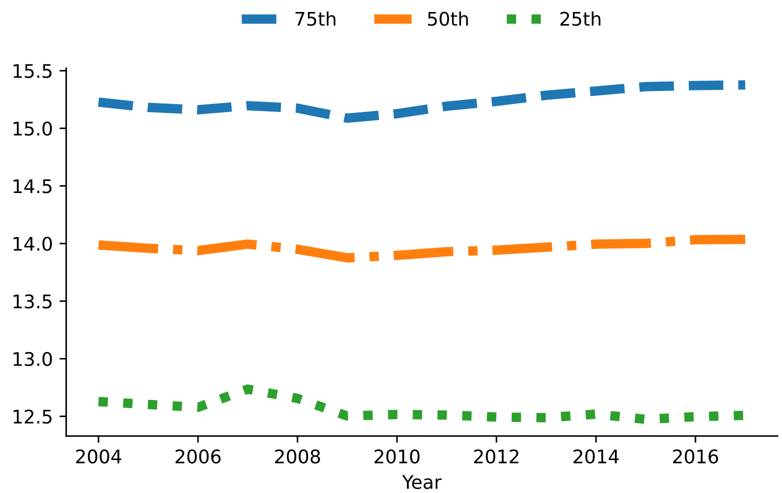


Notes: The figure shows the dispersion of log firm size by firm age as indicated in the legend for alternative size measures. The left panel uses intermediate inputs and applies the interquartile range as the dispersion measure. The middle panel uses the capital stock and shows the difference between the 95th and 50th percentile. The right panel uses employment and applies the standard deviation as the dispersion measure (restricting to positive values).

Figure A-6 shows firm-size dispersion conditional on age for the three alternative size measures. The increase in size dispersion conditional on age is even more visible when using intermediate inputs, and is again particularly pronounced for older firms: within-group size dispersion for the age groups 6–10, 11–15, and 16–20 increases by about 0.4 log points. The increase is even stronger when using the capital stock as the size measure: for the same age groups the increase exceeds 0.5 log points. This should, however, be interpreted with caution, as the measure focuses on dispersion in the right tail of the size distribution. Moreover, the capital stock captures only physical capital, so the increase in dispersion conditional on age is not driven by intangible assets. The last panel shows dispersion in employment conditional on age over time. Since this restriction heavily selects the sample, particularly among young firms, I only show the trends for older firm ages. For the age groups 11–15 and 16–20, a clear upward trend is noticeable. This increase is slightly smaller than the increase in the standard deviation of log sales shown in Figure A-2, though this should be interpreted with caution given sample selection.

Lastly, I show the size distribution conditional on age to examine how individual percentiles have shifted. Figure A-7 shows the 25th, 50th, and 75th percentile of the log sales distribution for 11–15 year old firms. The figure shows that the IQR increases due to changes in both the 25th and 75th percentile: the former decreased slightly while the latter rose.

Figure A-7: Size distribution (ages 11-15)



Notes: The figure shows percentiles (25th, 50th, 75th) of the log sales distribution for 11-15 year old firms.

C Model

This section lays out an extension of the Peters (2020) economy with ex-ante heterogeneity in firm productivity.

C.1 Preferences and aggregate economy

Time is continuous and indexed by t. The economy consists of a representative household that chooses the path of consumption C_t and wealth A_t to maximize lifetime utility

$$U = \int_0^\infty \exp(-\rho t) \ln C_t dt,$$

subject to the budget constraint $\dot{A}_t = r_t A_t + w_t L_t - C_t$. ρ denotes the discount rate, r_t the interest rate and w_t the real wage. The household supplies one unit of labor inelastically, i.e., $L_t = 1$. The optimality condition (Euler equation) for the household problem reads

$$\frac{\dot{C}_t}{C_t} = r_t - \rho.$$

Aggregate output is produced competitively using a Cobb-Douglas technology over a continuum of different products indexed by i (t subscripts are suppressed when convenient)

$$Y = \exp\left(\int_0^1 \ln\left[q_i y_i\right] di\right),$$

where y_i and q_i denote the quantity and quality of product i. Output is consumed entirely such that $Y = C$. Expenditure minimization leads to the standard demand function

$$y_i = \frac{YP}{p_i}. \tag{A-1}$$

P is defined as the aggregate price index, which I normalize to 1.

C.2 Production

Firms, indexed by f, produce in product market i with a linear technology

$$y_{ift} = \varphi_f l_{ift},$$

where y_{ift} denotes output, l_{ift} labor hired, and φ_f (time-invariant) firm productivity. Importantly, φ_f differs across firms, which captures the notion that some firms are systematically more efficient at producing than others in all of their product lines, e.g., due to a better business plan. As in Aghion et al. (2023), I assume two productivity types, i.e., $\varphi_f \in \{\varphi^h, \varphi^l\}$ where $\varphi^h/\varphi^l > 1$, which I refer to as high- and low-type firms.

C.3 Static allocation

Taking the joint distribution of product qualities and firm productivity as exogenous in this section, I characterize the static allocations at the product, firm and aggregate levels.

C.3.1 Product level

Firms in product market i compete in prices (Bertrand competition). In equilibrium, only the firm with the highest quality-adjusted productivity $q_{if}\varphi_f$ produces product i (henceforth, incumbent). Under Bertrand competition, the incumbent firm engages in limit pricing and sets its price equal to the quality-adjusted marginal costs of the follower (the firm with the second highest quality-adjusted productivity)

$$p_{if} = \frac{q_{if}}{q_{if'}} \frac{w}{\varphi_{f'}}, \tag{A-2}$$

where f' indexes the follower in product market i. The price that the incumbent sets is increasing in the quality gap between the incumbent and the follower, as eq. (A-2) shows. Defining the product markup as the output price over marginal costs, it follows

$$\mu_{if} \equiv \frac{p_{if}}{w/\varphi_f} = \frac{q_{if}}{q_{if'}} \frac{\varphi_f}{\varphi_{f'}}. \tag{A-3}$$

The incumbent's markup for product i is increasing in its quality and productivity gap. The price setting of the incumbent gives rise to the following profits for product i

$$\pi_{if} = p_{if} y_{if} - w l_{if} = Y \left(1 - \frac{1}{\mu_{if}}\right),$$

with labor demand for product i

$$l_{if} = \frac{Y}{w} \mu_{if}^{-1}. \tag{A-4}$$

Employment in product line i is decreasing in the markup.

C.3.2 Firm level

Firm employment is the sum of employment across the firm's product lines

$$l_f = \sum_{i \in N_f} l_{if} = \frac{Y}{w} \left(\sum_{i \in N_f} \mu_{if}^{-1} \right),$$

where N_f denotes the set of product lines in which firm f is the incumbent producer. Firm employment decreases in the markups within each product line but increases in the number of product lines. Hence, holding product markups constant, firms that produce in more product lines feature higher employment. Vice versa, holding the number of product lines constant, firms with higher product markups employ less labor. As sales are equalized across product lines, firm sales are given by $\left|N_f\right| Y \equiv n_f Y$, where n_f denotes the number of products firm f is producing. Hence, firms that produce in more product lines feature higher sales. Lastly, I define the firm markup μ_f as total firm sales over the wage bill $w l_f$.

C.3.3 Aggregate level

Integrating employment across firms or products yields the total workforce in production:

$$L_P = \int_f l_f df = \frac{Y}{w} \int_0^1 \mu_{if}^{-1} di. \tag{A-5}$$

Taking logs and integrating eq. (A-3), one obtains an expression for the wage

$$w = \exp\left(\int_0^1 \ln q_{if} di\right) \times \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right) \times \exp\left(\int_0^1 \ln \mu_{if}^{-1} di\right). \tag{A-6}$$

To find an expression for aggregate output, insert eq. (A-6) into eq. (A-5) to obtain

$$Y = Q\Phi\mathcal{M}L_P, \tag{A-7}$$

where

$$Q = \exp\left(\int_0^1 \ln q_i di\right), \quad \Phi = \exp\left(\int_0^1 \ln \varphi_{f(i)} di\right), \quad \mathcal{M} = \frac{\exp\left(\int_0^1 \ln \mu_i^{-1} di\right)}{\int_0^1 \mu_i^{-1} di}.$$

Aggregate output Y depends on geometric averages of quality Q and productivity Φ as well as on misallocation $\mathcal{M}$ and production labor L_P. Misallocation arises from markup dispersion ($\mathcal{M}$ is bounded by unity from above) that is due to quality and productivity heterogeneity. The product of Q, Φ and $\mathcal{M}$ captures aggregate Total Factor Productivity (TFP).

C.4 Dynamic firm problem

Firms compete for product markets through innovation (R&D). Via expansion R&D, a firm improves the quality of a randomly selected product operated by a competitor, thereby becoming the highest-quality producer in a product line that is new to the firm. Firms also undertake internal R&D, which enables them to further raise the quality of products they already control after successful expansion.

Item quality is improved step-wise such that every innovation (external or internal) increases q_i by a factor of λ.[11] Denoting by λ^{Δ_i} the relative qualities of incumbent and second-best firms within a product line, i.e.,

$$\lambda^{\Delta_i} = \frac{q_{if}}{q_{if\prime}}$$

[11] As Aghion et al. (2023), I assume that the step size of quality improvements exceeds the productivity differential, $\lambda > \varphi^h/\varphi^l$. This assumption ensures that the firm with the highest quality version in a product line is the incumbent producer. Relaxing this assumption would give room for a race for incumbency between low-productivity entrants facing a high-productivity incumbent from which I abstract.

and by $[\mu_i]$ a firm's set of markups across its product lines, firm profits follow

$$\pi_f(n,[\mu_i]) \equiv \sum_{k=1}^{n} \pi_i(\mu_k) = \sum_{k=1}^{n} Y\left(1-\frac{1}{\mu_k}\right) = \sum_{k=1}^{n} Y\left(1-\frac{1}{\lambda^{\Delta_k}}\frac{1}{\frac{\varphi_{f(k)}}{\varphi_{f'(k)}}}\right).$$

Incumbent firms choose the rate of expansion R&D, x_{it}, and the rate of internal R&D, I_{it}, for each of their product lines, i. When choosing x_{it} and I_{it}, firms take aggregate output Y_t, the real wage w_t, the share of lines operated by high-productivity firms S_t, the interest rate r_t and the rate of creative destruction τ_t as given. Note that S_t is predetermined in t and its evolution is characterized shortly. Denoting the time derivative by $\dot{V}_t^h()$, the value function of a high-productivity type firm (indexed by h) satisfies the following HJB equation:

$$\begin{aligned}
&r_t V_t^h(n,[\mu_i],S_t) - \dot{V}_t^h(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \underbrace{\pi_i(\mu_k)}_{\text{Flow profits}} + \sum_{k=1}^{n} \underbrace{\tau_t\Big[V_t^h\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i],S_t)\Big]}_{\text{Creative destruction}} \\
&\quad + \max_{[I_{it},x_{it}]}\left\{ \sum_{k=1}^{n} \underbrace{I_{kt}\Big[V_t^h\left(n,\left[[\mu_i]_{i\neq k},\mu_k\cdot\lambda\right],S_t\right) - V_t^h(n,[\mu_i],S_t)\Big]}_{\text{Internal R\&D}} \right. \\
&\quad + \sum_{k=1}^{n} \underbrace{x_{kt}\Big[S_t V_t^h\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) + (1-S_t)V_t^h\left(n+1,\left[[\mu_i],\lambda\cdot\frac{\varphi^h}{\varphi^l}\right],S_t\right) - V_t^h(n,[\mu_i],S_t)\Big]}_{\text{Expansion R\&D}} \\
&\quad \left. - \underbrace{w_t\Gamma\left([x_{it},I_{it}];n,[\mu_i]\right)}_{\text{R\&D costs}} \right\}.
\end{aligned}$$

The value of a firm consists of flow profits, research costs, and three parts related to internal R&D, expansion R&D, and creative destruction. At the rate of creative destruction τ_t (determined in equilibrium), the firm loses one of its n products, in which case, it remains with $n-1$ products. At the optimally chosen rate I_{it}, internal R&D turns out successful (third row), and the firm charges a λ times higher markup on its product according to eq. (A-3). Alternatively, at the optimally chosen rate x_{it}, expansion R&D is successful (fourth row), and the firm acquires a new product (n increases by one).

Productivity-type heterogeneity introduces novel elements to the value function of the firm. The share of product lines operated by each productivity type S_t is a state variable that firms keep track of to build markup expectations. When taking over a new product line through expansion R&D (fourth row), the probability of replacing a high-type incumbent is S_t, in which case the high-type entrant charges a markup of λ. With probability $1-S_t$, the replaced incumbent is of the low type, and the high-type entrant charges a markup of $\lambda \cdot \varphi^h/\varphi^l$. Firms take S_t as given; however, they affect it through their expansion R&D efforts x_{it} in equilibrium. The HJB equation for a low-productivity firm follows the same structure and is listed in the Appendix, Section D.1. The term related to expansion R&D (fourth row) is distinct since markup expectations differ for low-productivity firms.

$\Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right)$ denote the R&D costs. For their R&D activities, firms pay a cost of

$$\Gamma\left([x_{it}, I_{it}]; n, [\mu_i]\right) = \sum_{k=1}^{n} c\left(x_{kt}, I_{kt}; \mu_k\right) = \sum_{k=1}^{n} \left[\mu_k^{-1} \frac{1}{\psi_I} (I_{kt})^{\zeta} + \frac{1}{\psi_x} (x_{kt})^{\zeta}\right]$$

in terms of labor. $\zeta > 1$ ensures convexity of the cost function. R&D costs are additively separable to render a closed-form solution of the value function along the balanced growth path. ψ_I and ψ_x scale internal and external R&D costs and capture the R&D efficiency.[12]

The rate of creative destruction τ_t that firms take as given in their optimization problem is, in equilibrium, equal to the sum of expansion R&D rates and the rate of entry z_t

$$\tau_t = \int_0^1 x_{it} di + z_t. \tag{A-8}$$

Firm entry is determined as follows. Potential entrants produce a flow rate of entry z_t using a technology that is linear in labor: $z_t = \psi_z L_{Et}$, where ψ_z denotes the entry efficiency and L_{Et} research labor of entrants. Entrants improve the quality of a randomly selected product line. The productivity type is realized after entry and assigned with the exogenous probabilities p^h and $1 - p^h$, respectively. Entrants start with a one-step quality gap. When $z_t > 0$, the free entry condition requires that the expected value of firm entry equals the entry costs

$$p^h E[V_t^h(1, \mu_i)] + (1 - p^h) E[V_t^l(1, \mu_i)] = \frac{1}{\psi_z} w_t, \tag{A-9}$$

where the expected value of entering as a high- or low-type firm is

$$\begin{aligned} E[V_t^h(1, \mu_i)] =& S_t V_t^h(1, \lambda) + (1 - S_t) V_t^h(1, \lambda \times \varphi^h/\varphi^l) \\ E[V_t^l(1, \mu_i)] =& S_t V_t^l(1, \lambda \times \varphi^l/\varphi^h) + (1 - S_t) V_t^l(1, \lambda). \end{aligned}$$

Firms' expansion and internal R&D efforts generate a two-dimensional distribution of quality

[12] I follow Peters (2020) and scale internal R&D costs by the inverse markup, which removes the dependence of internal R&D on product-line markups while preserving its sensitivity to all other equilibrium forces.

and productivity gaps across product lines, denoted by $\nu_t(\Delta, \frac{\varphi_f}{\varphi_{f'}})$. These gaps determine product markups, so the distribution ν_t characterizes the overall markup distribution in the economy. I derive ν_t as a function of firm policies in Appendix D.2. Importantly, the marginal distributions of ν_t determine S_t, the share of product lines operated by high-productivity firms, which firms take as given in their optimization problem. The law-of-motion for S_t follows from ν_t:

$$\dot{S}_t = \sum_{i=1}^{\infty} \left[\dot{\nu}_t \left(i, \frac{\varphi^h}{\varphi^h} \right) + \dot{\nu}_t \left(i, \frac{\varphi^h}{\varphi^l} \right) \right] = S_t x_t^h + z_t p^h - S_t \tau_t. \tag{A-10}$$

Hence, S_t increases through the expansion of high-type incumbents into new product lines and the entry of new high-type firms, and decreases due to creative destruction.

Lastly, labor market clearing requires that production labor L_{Pt} and total research labor L_{Rt} add up to one, the aggregate labor endowment

$$1 = L_{Pt} + L_{Rt} = \frac{Y_t}{w_t} \int_0^1 \mu_i^{-1} di + \int_0^1 \left(\mu_i^{-1} \frac{I_{it}^{\zeta}}{\psi_I} + \frac{x_{it}^{\zeta}}{\psi_x} \right) di + \frac{z_t}{\psi_z}. \tag{A-11}$$

C.5 Balanced growth path characterization

I define a balanced growth path of the economy as follows.

Definition 1. *A balanced growth path (BGP) is a set of allocations $[x_{it}, I_{it}, \ell_{it}, z_t, S_t, y_{it}, C_t]_{it}$ and prices $[r_t, w_t, p_{it}]_{it}$ such that firms choose $[x_{it}, I_{it}, p_{it}]$ optimally, the representative household maximizes utility choosing $[C_t, y_{it}]_{it}$, the growth rate of aggregate variables is constant, the free-entry condition holds, all markets clear and the distribution of quality and productivity gaps ν_t is stationary.*

Along the balanced growth path, the economy can be characterized in closed form.

Proposition A-1. *In the above setup, along a balanced growth path:*

1. *The value of a product line i for a firm of productivity type $d \in \{h, l\}$ is given by*

$$V_i^d(1, \mu_i, S) = \frac{1}{\rho + \tau} \left[Y_t \left(1 - \frac{1}{\mu_i} \right) + \frac{\zeta - 1}{\psi_x} x_i^{\zeta} w_t + \frac{\zeta - 1}{\psi_I} I_i^{\zeta} w_t \mu_i^{-1} \right] \tag{A-12}$$

   The value of a firm is the sum of its product line values $V^d(n, [\mu_i], S) = \sum_{k=1}^{n} V_i^d(1, \mu_k, S)$.

2. *Expansion R&D rates are firm productivity-type specific, i.e., $x_i \in \{x^h, x^l\}$. Further, high-productivity firms expand faster than low-productivity ones $x^h > x^l$.*

3. The share of product lines operated by high-productivity type firms S is determined by

$$S\tau = Sx^h + zp^h. \tag{A-13}$$

4. The growth rate of aggregate variables is given by

$$g = \frac{\dot{Q}_t}{Q_t} = \Bigg(\underbrace{I}_{\textit{Incumbent internal R\&D}} + \underbrace{Sx^h + (1-S)x^l}_{\textit{Incumbent expansion R\&D}} + \underbrace{z}_{\textit{Entry}} \Bigg) \times \ln(\lambda). \tag{A-14}$$

Proof. The Appendix, Sections D.1, D.2 and D.3 contain the proofs. □

The value of a product line in eq. (A-12) consists of three terms: profits for a given markup and the continuation values of expansion and internal R&D. The sum of the three terms is discounted by the discount rate and the rate of creative destruction. The more impatient the household or the higher the risk of replacement, the lower the value of a product line. Importantly, expansion R&D rates are productivity-type specific. More productive incumbents charge higher markups, which increases the expected value of a product line. The optimality condition for expansion R&D (Appendix, eq. A-30) equates the expected value of a product line with the marginal cost of expansion R&D. Hence, in equilibrium, more productive firms pay a higher marginal cost of expansion R&D and choose $x^h > x^l$. By contrast, internal R&D rates are type-invariant, $I_i \equiv I$. Because expansion rates differ by type, the values of a product line and a firm are productivity-type specific as well.

Proposition 1 further shows that the share of product lines operated by high-productivity type firms is constant along the balanced growth path. The expression in eq. (A-13) directly follows by setting the law-of-motion for S_t in eq. (A-10) equal to zero. Eq. (A-13) captures that, along the balanced growth path, the share of product lines by high-productivity firms that fall victim to creative destruction is exactly equal to the share of product lines that high-productivity firms gain by entering new product lines through incumbent creative destruction or firm entry. Eq. (A-13) can further be rearranged to

$$S = \frac{zp^h}{(1-S)(x^l - x^h) + z}, \tag{A-15}$$

which shows that S depends on the difference in the expansion R&D rates between firm types, $x^l - x^h$. Holding firm entry z fixed, an increase in the expansion rate of high-productivity incumbents must be matched by an equal rise in the expansion rate of less-productive firms for S to remain constant. Note that positive firm entry is necessary for both firm types to co-exist in equilibrium where $x^l \neq x^h$ and S is constant at its interior solution.

Long-run growth results from R&D at the product level. This occurs through successful internal R&D or (incumbents' and entrants') creative destruction. The aggregate arrival rate of innovation is, hence, equal to the sum of the rates of creative destruction τ and internal R&D I. Multiplying the arrival rate by the log step size of innovation delivers the aggregate growth rate g, as shown in eq. (A-14) of Proposition A-1. Since expansion R&D rates are heterogeneous, i.e., $x^h > x^l$, changes in the share of product lines operated by each productivity type, S and $1-S$, affect the aggregate growth rate. Along the balanced growth path, both τ and g are constant. I further characterize the aggregate labor income share, the TFP misallocation measure $\mathcal{M}$, and the aggregate markup analytically in the Appendix, Section D.2.

To find the balanced growth path, I jointly solve the optimality conditions of the firm (derived in Appendix D.1), the free entry condition, eq. (A-9), the labor market clearing condition, eq. (A-11), and the system of differential equations characterizing the distribution of productivity and quality gaps captured in eqs. (A-32) and (A-33). Appendix D.4 contains the details.

C.6 Firm dynamics

Firms lose products according to the same stochastic process as in Klette and Kortum (2004). However, in this model, firms add products at systematically different rates as optimally chosen expansion R&D rates vary with the firm's productivity type.[13] The following section derives productivity-type specific firm-size dynamics and survival probabilities.

C.6.1 Sales-age relationship

Firm sales are proportional to the number of products a firm produces. As such, successful expansion R&D increases firm sales. Since optimal expansion R&D rates are productivity-type specific, so is the sales-age relationship. For productivity type φ_f, the average log sales of surviving firms at age a_f relative to the average log sales of entrants is

$$E\left[\ln n_f Y | a_f, \varphi_f\right] - E\left[\ln n_f Y | 0, \varphi_f\right] = \underbrace{E\left[\ln n_f | a_f, \varphi_f\right]}_{\text{Avg. product count}},$$

[13]Therefore, the properties related to firm-size growth and survival in Klette and Kortum (2004) hold conditional on the firm type. In particular, conditional on the type, firm size, and growth are unrelated, as in Lentz and Mortensen (2008). For the unconditional firm-size and growth correlation, two forces are at play. On the one hand, young (small) firms tend to grow quicker due to the survival bias. On the other hand, more productive firms (with faster growth rates) are more likely to end up large.

where n_f is a firm's number of products. The probability of producing n products at age a conditional on survival is $(1-\gamma^j(a))\left(\gamma^j(a)\right)^{n-1}$, where $\gamma^j(a) = x^j \frac{1-e^{-(\tau - x^j)a}}{\tau - x^j e^{-(\tau-x^j)a}}$ and $j \in \{h, l\}$. Therefore, the type-specific sales-age relationship can be reformulated as

$$E\left[\ln n_f Y | a_f, \varphi_f\right] - E\left[\ln n_f Y | 0, \varphi_f\right] = \underbrace{\left(1-\gamma^f(a_f)\right)\sum_{n=1}^{\infty} \ln n \times \left(\gamma^f(a_f)\right)^{n-1}}_{\text{Avg. product count}}. \tag{A-16}$$

C.6.2 Employment-age relationship

Average employment conditional on age and productivity type is equal to

$$E[\ln l_f | a_f, \varphi_f] = \ln\left(\frac{Y}{w}\right) + E\left[\ln n_f | a_f, \varphi_f\right] - E\left[\ln \mu_f | a_f, \varphi_f\right].$$

Since $\frac{Y}{w}$ is constant, the type-specific employment-age relationship simplifies to

$$E[\ln l_f | a_f, \varphi_f] - E[\ln l_f | 0, \varphi_f] = \underbrace{E\left[\ln n_f | a_f, \varphi_f\right]}_{\text{Avg. product count}} - \underbrace{\left(E\left[\ln \mu_f | a_f, \varphi_f\right] - E\left[\ln \mu_f | 0, \varphi_f\right]\right)}_{\text{Markup-age relationship}}, \tag{A-17}$$

where $E\left[\ln n_f | a_f, \varphi_f\right]$ is defined in eq. (A-16) and the markup-age relationship, $E\left[\ln \mu_f | a_f, \varphi_f\right] - E\left[\ln \mu_f | 0, \varphi_f\right]$, is derived in Appendix D.6.

C.6.3 Firm survival

Firms that lose their last product exit the economy. Since firm expansion is type-dependent, so is firm survival. The survival function in Klette and Kortum (2004) holds conditional on the firm type, i.e., the share of high and low type firms surviving until age a_f is

$$\chi^h(a_f) = 1 - \tau \frac{1-e^{-(\tau - x^h)a_f}}{\tau - x^h e^{-(\tau - x^h)a_f}} \quad \text{and} \quad \chi^l(a_f) = 1 - \tau \frac{1-e^{-(\tau - x^l)a_f}}{\tau - x^l e^{-(\tau - x^l)a_f}}. \tag{A-18}$$

Hence, the share of high-type firms among surviving firms at age a_f is

$$s^h(a_f) = \frac{p^h \chi^h(a_f)}{p^h \chi^h(a_f) + (1-p^h)\chi^l(a_f)}, \tag{A-19}$$

which corresponds to the mass of high-type survivors relative to the total mass of survivors. Since $x^h > x^l$, it is easy to show that the sales-age relationship of high-productivity firms is steeper than the one of low-productivity ones and that the share of the former among surviving firms $s^h(a_f)$ increases in a_f.

C.6.4 Components of the size dispersion decomposition

Mean size by age Mean log size conditional on age pools both productivity types with the type-share weights from eq. (A-19):

$$\mu(a_f) \equiv E\left[\ln \text{Size}_{f,t} \mid \text{Age}_{f,t} = a_f\right] = s^h(a_f) \cdot \mu^h(a_f) + \left(1 - s^h(a_f)\right) \cdot \mu^l(a_f), \tag{A-20}$$

where $\mu^d(a_f) = E[\ln \text{Size}_{f,t}|\text{Age}_{f,t} = a_f, \varphi_f = \varphi^d]$ is the expected log product count from eq. (A-16). Since all entrants start with a single product line, $\mu^d(0) = \ln(1) = 0$ for both types, so $\mu(a_f)$ coincides with size growth relative to entry. More generally, the between-age component of the decomposition subtracts the grand mean $\bar{\mu} = \sum_a p_a \mu(a)$, so the decomposition is invariant to any common additive shift in μ. Mean size rises with age because type-specific product counts $\mu^d(a_f)$ grow as firms accumulate product lines, and the share $s^h(a_f)$ shifts toward the type with the steeper size-age trajectory.

Age distribution Since each entrant independently draws a high-type productivity with probability p^h and a low-type productivity with probability $1-p^h$, the unconditional survival probability to age a_f is

$$\chi(a_f) = p^h \chi^h(a_f) + (1 - p^h)\chi^l(a_f), \tag{A-21}$$

where χ^h and χ^l are defined in eq. (A-18). In the numerical implementation of the decomposition, continuous age is approximated on a fine grid with step size Δa and midpoints $\{a_k\}$. The discrete probability mass assigned to each age bin is

$$\omega(a_k) = \frac{p^h \chi^h(a_k) + (1 - p^h)\chi^l(a_k)}{\sum_j \left[p^h \chi^h(a_j) + (1 - p^h)\chi^l(a_j)\right]}.$$

Size dispersion conditional on age Within any age cohort, log-size dispersion arises from two sources: the allocation of firms across productivity types and the stochastic accumulation of product lines. The number of product lines of a surviving firm of type d at age a_f follows a geometric distribution with parameter $\gamma^d(a_f)$ (see eq. (A-16)):

$$P(n_f = n \mid a_f,\, \varphi_f = \varphi^d) = \left(1 - \gamma^d(a_f)\right)\left(\gamma^d(a_f)\right)^{n-1}, \quad n = 1, 2, \ldots$$

Pooling both types with weight $s^h(a_f)$ from eq. (A-19), the mixture distribution of product count conditional on age is

$$P(n_f = n \mid a_f) = s^h(a_f) \cdot \Big(1 - \gamma^h(a_f)\Big)\Big(\gamma^h(a_f)\Big)^{n-1} + \Big(1 - s^h(a_f)\Big) \cdot \Big(1 - \gamma^l(a_f)\Big)\Big(\gamma^l(a_f)\Big)^{n-1}.$$

The variance of log sales conditional on age a_f therefore equals the variance of log product count:

$$\mathrm{Var}(\ln \mathrm{Size}_{f,t} \mid a_f) = \sum_{n=1}^{\infty} P(n_f = n \mid a_f) \cdot (\ln n - \bar{\mu}(a_f))^2 , \tag{A-22}$$

where $\bar{\mu}(a_f) = \sum_{n=1}^{\infty} \ln(n) \cdot P(n_f = n \mid a_f)$ is the mean log product count.

Implementation with integer age groupings Empirically, firm age is observed at annual frequency. For firms of integer age a (i.e., $a_f \in [a, a+1)$), the law of total variance applied across exact continuous ages within the bin gives within-bin size variance

$$V_a = \underbrace{\sum_{k:\, a_k \in [a,a+1)} \frac{\omega(a_k)}{p_a} \cdot \mathrm{Var}(\ln n_f \mid a_k)}_{\text{avg. within-exact-age variance}} + \underbrace{\sum_{k:\, a_k \in [a,a+1)} \frac{\omega(a_k)}{p_a} \cdot (\mu(a_k) - \bar{\mu}_a)^2}_{\text{variance of mean size across exact ages}}, \tag{A-23}$$

where $p_a = \sum_{k:\, a_k \in [a,a+1)} \omega(a_k)$ is the mass of firms in age bin a, $\mu(a_k)$ is the mean log size at fine-grid midpoint a_k from eq. (A-20), and $\bar{\mu}_a = \sum_{k:\, a_k \in [a,a+1)} (\omega(a_k)/p_a) \cdot \mu(a_k)$ is the within-bin conditional mean. The total cross-sectional variance then decomposes as

$$\mathrm{Var}(\ln \mathrm{Size}) = \underbrace{\sum_a p_a \cdot V_a}_{\text{within-age}} + \underbrace{\sum_a p_a \cdot (\bar{\mu}_a - \bar{\mu})^2}_{\text{between-age}},$$

where $\bar{\mu} = \sum_a p_a \bar{\mu}_a$ is the grand mean log size.

C.6.5 Firm size distribution

The theory makes precise predictions about the firm-size distribution. I derive the firm-size distribution in Section D.7 in the Appendix. Denoting by M^h the mass of high-productivity type firms and by M the total mass of firms, the share of high-productivity type firms in the cross section is

$$S_{M^h} = \frac{M^h}{M}, \tag{A-24}$$

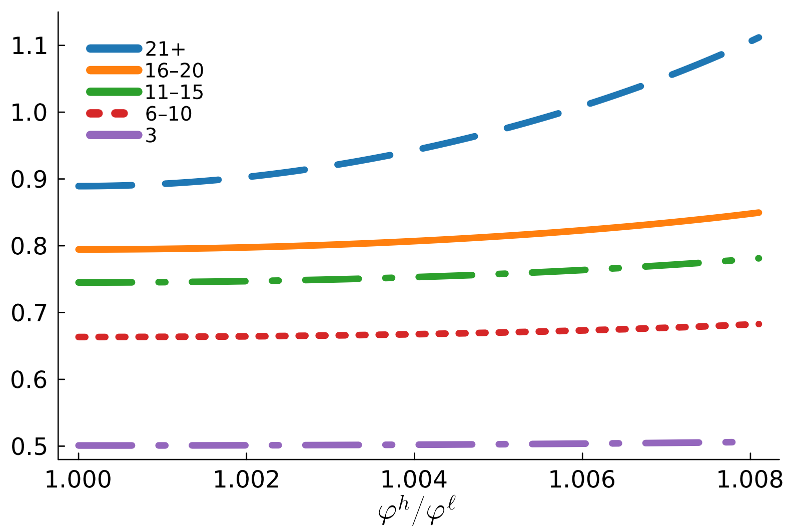
and the firm entry rate equals

$$\text{Firm entry rate} = \frac{z}{M}. \tag{A-25}$$

C.7 Comparative statics

To confirm that productivity divergence can generate the observed empirical patterns, I run comparative statics on the productivity differential, starting from the calibrated Peters (2020) economy with $\varphi^h/\varphi^\ell = 1$.[14] Figure A-8 shows size dispersion conditional on age for different values of φ^h/φ^ℓ. As the measure of size dispersion I use the standard deviation of log sales. First, note that dispersion levels undershoot their data equivalent shown in Figure A-2. This is because there is too little within-age dispersion in Peters (2020) as discussed previously in the main text. Second, even a modest productivity gap of 0.8% generates substantial within-age dispersion. Among firms aged 21 and older, within-age dispersion rises by 0.2 log points, while among young firms it is nearly flat. This age-dependent increase directly mirrors the empirical pattern in Figure A-2.

Figure A-8: Firm-size dispersion conditional on age (model)



Notes: The figure shows the standard deviation of log sales by firm age indicated in the legend. φ^h/φ^ℓ denotes the productivity gap between high- and low-productivity firms.

D Model details

[14] I assume that entrants are assigned either productivity type with equal probability, i.e., $p^h = 0.5$. This exercise is a simple mechanism illustration.

D.1 Solving the dynamic firm problem

The HJB for a high productivity-type firm h reads[15]

$$
\begin{aligned}
&r_t V_t^h(n,[\mu_i],S_t) - \dot{V}_t^h(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^h\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \left\{ \sum_{k=1}^{n} I_k \Big[V_t^h\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \right. \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^h\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) + (1-S_t) V_t^h\left(n+1,\left[[\mu_i],\lambda\times\varphi^h/\varphi^l\right],S_t\right) - V_t^h(n,[\mu_i],S_t) \Big] \\
&\quad \left. - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \right\}
\end{aligned}
$$

The HBJ for a low productivity-type firm l reads

$$
\begin{aligned}
&r_t V_t^l(n,[\mu_i],S_t) - \dot{V}_t^l(n,[\mu_i],S_t) = \\
&\quad \sum_{k=1}^{n} \pi(\mu_k) + \sum_{k=1}^{n} \tau_t \Big[V_t^l\left(n-1,[\mu_i]_{i\neq k},S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad + \max_{[x_k,I_k]} \left\{ \sum_{k=1}^{n} I_k \Big[V_t^l\left(n,\left[[\mu_i]_{i\neq k},\mu_k\times\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \right. \\
&\quad + \sum_{k=1}^{n} x_k \Big[S_t V_t^l\left(n+1,\left[[\mu_i],\lambda\times\varphi^l/\varphi^h\right],S_t\right) + (1-S_t) V_t^l\left(n+1,\left[[\mu_i],\lambda\right],S_t\right) - V_t^l(n,[\mu_i],S_t) \Big] \\
&\quad \left. - w_t \left[\mu_k^{-1}\frac{1}{\psi_I}(I_k)^\zeta + \frac{1}{\psi_x}(x_k)^\zeta \right] \right\}.
\end{aligned}
$$

I solve for the value function of a high-type firm, however the steps for the low-type firm are equivalent. For clarity, I suppress the dependence of the value function on S_t in the following. Guess that the value function of the firm consists of a component that is common to all lines and a line-specific component

$$
V_t^h(n,[\mu_i]) = V_{t,P}^h(n) + \sum_{k=1}^{n} V_{t,M}^h(\mu_k).
$$

Substituting the guess into the HJB, $V_{t,P}^h(n)$ and $V_{t,M}^h(\mu_k)$ solve the following differential

[15] The derivation follows Peters (2020) when applicable.

equations

$$r_t V^h_{t,M}(\mu_i) - \dot{V}^h_{t,M}(\mu_i) = \pi(\mu_i) - \tau_t V^h_{t,M}(\mu_i) + \max_{I_i} \left\{ I_i \Big[V^h_{t,M}(\mu_i \times \lambda) - V^h_{t,M}(\mu_i) \Big] - w_t \mu_i^{-1} \frac{1}{\psi_I} (I_i)^\zeta \right\} \tag{A-26}$$

and

$$r_t V^h_{t,P}(n) - \dot{V}^h_{t,P}(n) = \sum_{k=1}^{n} \tau_t \Big[V^h_{t,P}(n-1) - V^h_{t,P}(n) \Big] + \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \Big[V^h_{t,P}(n+1) - V^h_{t,P}(n) + S_t V^h_{t,M}(\lambda) + (1 - S_t) V^h_{t,M}(\lambda \times \varphi^h / \varphi^l) \Big] - w_t \frac{1}{\psi_x} (x_k)^\zeta \right\}. \tag{A-27}$$

Assume that in steady-state $V^h_{t,P}$ and $V^h_{t,M}$ grow at the constant rate g. Using this guess in eq. (A-26) and following Peters (2020), we obtain for $V^h_{t,M}(\mu_i)$

$$V^h_{t,M}(\mu_i) = \frac{\pi(\mu_i) + \frac{\zeta - 1}{\psi_I} (I_i)^\zeta w_t \mu_i^{-1}}{\rho + \tau},$$

where I_i solves

$$I_i = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta - 1}{\psi_I} (I_i)^\zeta \right) \left(1 - \frac{1}{\lambda} \right) \frac{\psi_I}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}. \tag{A-28}$$

Eq. (A-28) shows that internal innovation rates I_i are time invariant, and independent of the product line and the productivity type of the firm, $I \equiv I^h = I^l$.

With this at hand, we can turn back to the differential equation for $V^h_{t,P}(n)$ in eq. (A-27). In addition to the guess that $V^h_{t,P}(n)$ grows at rate g, conjecture that $V^h_{t,P}(n) = n \times v^h_t$. Combined with the Euler we get

$$(\rho + \tau) n v^h_t = \max_{[x_k]} \left\{ \sum_{k=1}^{n} x_k \Big[v^h_t + S_t V^h_{t,M}(\lambda) + (1 - S_t) V^h_{t,M}(\lambda \times \varphi^h / \varphi^l) \Big] - w_t \frac{1}{\psi_x} (x_k)^\zeta \right\}. \tag{A-29}$$

The optimality condition for x_k is given by

$$v^h_t + S_t V^h_{t,M}(\lambda) + (1 - S_t) V^h_{t,M}(\lambda \times \varphi^h / \varphi^l) = w_t \frac{\zeta}{\psi_x} (x_k)^{\zeta - 1}. \tag{A-30}$$

Several observations are noteworthy. First, eq. (A-30) shows that optimal expansion rates are independent of quality and productivity gaps in line k. We can hence drop the item indexation: $x_k = x^d$, where $d \in \{h, \ell\}$. Second, $v_t, V^h_{t,M}, w_t$ all grow at the same rate g, which implies that expansion rates are constant over time. We can hence write eq. (A-29) as

$$v^h_t = \frac{1}{(\rho+\tau)} \frac{\zeta-1}{\psi_x} (x^h)^\zeta w_t.$$

Gathering all terms, the value function is given by

$$\begin{aligned} V^h_t(n, [\mu_i]) =& V^h_{t,P}(n) + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n v^h_t + \sum_{k=1}^{n} V_{t,M}(\mu_k) \\ =& n \frac{1}{(\rho+\tau)} \frac{\zeta-1}{\psi_x} (x^h)^\zeta w_t + \sum_{k=1}^{n} \frac{\pi(\mu_k) + \frac{\zeta-1}{\psi_I} I^\zeta w_t \mu_k^{-1}}{\rho+\tau}, \end{aligned} \tag{A-31}$$

which is the expression for the value function stated in the main text, Proposition A-1.

Using the expression for v^h_t, write the optimality condition in eq. (A-30) as

$$\begin{aligned} \frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t \left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \right) + (1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^l}{\varphi^h} \right) \\ = (\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1}. \end{aligned}$$

Following the same steps for low-productivity firms, we obtain the optimality condition

$$\begin{aligned} \frac{\zeta-1}{\psi_x}(x^l)^\zeta + S_t \left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^h}{\varphi^l} \right) + (1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\lambda}\right) + \frac{\zeta-1}{\psi_I} I^\zeta \lambda^{-1} \right) \\ = (\rho+\tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta-1}. \end{aligned}$$

Lastly, I prove that more productive firms choose higher expansion R&D rates, i.e., $x^h > x^l$. Intuitively, the proof shows that an increase in productivity raises the stream of profits in a product line. The continuation value and the marginal cost of expansion R&D have to rise for the optimality condition of the expansion R&D rate to hold, implying that the expansion R&D rate increases. First note that product markups are increasing in firm productivity, as shown in eq. (A-3). Next, I totally differentiate the optimality condition for the expansion R&D rate and show that the expansion R&D rate is increasing in the markup.

Write the optimality condition for expansion R&D as[16]

$$\frac{\zeta-1}{\psi_x}(x^h)^\zeta + S_t\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^h}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^h}\right)+(1-S_t)\left(\frac{Y_t}{w_t}\left(1-\frac{1}{\mu^l}\right)+\frac{\zeta-1}{\psi_I}I^\zeta\frac{1}{\mu^l}\right)$$
$$=(\rho+\tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta-1},$$

where μ^h and μ^l denote the initial markup charged when facing a high- and low-productivity firm. Totally differentiate with respect to markups and the expansion R&D rate

$$S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left((\rho+\tau)(x^h)^{\zeta-2}-(x^h)^{\zeta-1}\right)dx^h,$$

where $d\mu^h = d\mu^l \equiv d\mu$. Since $x^h > 0$, the above can be rearranged to

$$\frac{1}{(x^h)^{\zeta-2}}\left[S\frac{1}{(\mu^h)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)+(1-S)\frac{1}{(\mu^l)^2}\left(\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta\right)\right]d\mu$$
$$=\frac{\zeta(\zeta-1)}{\psi_x}\left(\rho+\tau-x^h\right)dx^h.$$

The left hand side captures the effect of changes in markups on profits. The right hand side captures the effect of changes in the expansion R&D rate on the continuation value and marginal costs of expansion R&D. From eq. (A-28) we know that $\frac{Y_t}{w_t}-\frac{\zeta-1}{\psi_I}I^\zeta>0$, otherwise the optimal internal R&D rate is negative. For a stationary firm size distribution, we must further have $\tau > x^h$. From this, it follows that $dx^h/d\mu > 0$, which concludes the proof.

D.2 Joint distribution of quality and productivity gaps

The distribution of quality and productivity gaps between incumbent and laggard firms characterizes economic aggregates in the model. On the one hand, quality and productivity gaps define product markups that determine labor demand. On the other hand, the joint distribution characterizes the share of product lines operated by each productivity type, which is a state variable in the firm's optimization problem. This section derives the joint distribution of quality and productivity gaps as a function of firm policies, which allows the equilibrium distribution to be solved jointly with the policies.

The distribution of quality and productivity gaps across product lines is characterized by

[16] This uses the optimality condition of the high-type firm but the one of the low type works equivalently.

a set of infinitely many differential equations. For simplicity, I characterize the differential equations for firm-type specific expansion R&D rates, x_t^h and x_t^ℓ, and homogeneous internal R&D rates, I_t, as proven in Proposition A-1 for a balanced growth path.[17] The measure ν_t of product lines in which an incumbent of productivity type φ_f is Δ quality steps ahead of a $\varphi_{f'}$ type laggard follows (for $\Delta \geq 2$)

$$\dot{\nu}_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right) = I_t\nu_t\left(\Delta - 1, \frac{\varphi_f}{\varphi_{f'}}\right) - \nu_t\left(\Delta, \frac{\varphi_f}{\varphi_{f'}}\right)(I_t + \tau_t). \tag{A-32}$$

For product lines where the incumbent is one step ahead ($\Delta = 1$), this measure follows

$$\begin{aligned}
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^h}\right) &= (1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t - \nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)(I_t + \tau_t)\\
\dot{\nu}_t\left(1, \frac{\varphi^l}{\varphi^l}\right) &= (1 - S_t)x_t^l(1 - S_t) + z_t(1 - p^h)(1 - S_t) - \nu_t\left(1, \frac{\varphi^l}{\varphi^l}\right)(I_t + \tau_t)\\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^h}\right) &= S_t x_t^h S_t + z_t p^h S_t - \nu_t\left(1, \frac{\varphi^h}{\varphi^h}\right)(I_t + \tau_t)\\
\dot{\nu}_t\left(1, \frac{\varphi^h}{\varphi^l}\right) &= S_t x_t^h(1 - S_t) + z_t p^h(1 - S_t) - \nu_t\left(1, \frac{\varphi^h}{\varphi^l}\right)(I_t + \tau_t).
\end{aligned} \tag{A-33}$$

Changes in ν_t are due to inflows and outflows. Outflows from state Δ arise from successful internal R&D (quality gap increases from Δ to $\Delta + 1$) and creative destruction (Δ is reset to unity), as captured by the terms following the minus signs. Inflows vary with the quality gap. For $\Delta \geq 2$, inflows into state Δ are due to successful internal R&D in product lines with quality gaps of $\Delta - 1$. For $\Delta = 1$, inflows result from creative destruction. For example, the measure of products with a low-type incumbent and high-type laggard $\nu_t\left(1, \frac{\varphi^l}{\varphi^h}\right)$ increases due to low-type incumbents and entrants replacing high-type incumbents, captured by $(1 - S_t)x_t^l S_t + z_t(1 - p^h)S_t$ in eq. (A-33).

The following characterizes the two-dimensional distribution of quality and productivity gaps along the BGP. I solve for the steady state distribution over quality and productivity gaps by setting the differential equations characterizing the law-of-motion in eq. (A-32) and (A-33) equal to zero. From this, one obtains the stationary mass of product lines with quality gap

[17] The distribution along the transition path is characterized in the Appendix, Section **??**.

λ^{Δ} and productivity gap φ^{i}/φ^{j}

$$\begin{aligned}
\nu\left(\Delta,\frac{\varphi^{l}}{\varphi^{h}}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta}\frac{(1-S)x^{l}S+z(1-p^{h})S}{I}\\
\nu\left(\Delta,\frac{\varphi^{l}}{\varphi^{l}}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta}\frac{(1-S)x^{l}(1-S)+z(1-p^{h})(1-S)}{I}\\
\nu\left(\Delta,\frac{\varphi^{h}}{\varphi^{h}}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta}\frac{Sx^{h}S+zp^{h}S}{I}\\
\nu\left(\Delta,\frac{\varphi^{h}}{\varphi^{l}}\right) &= \left(\frac{I}{I+\tau}\right)^{\Delta}\frac{Sx^{h}(1-S)+zp^{h}(1-S)}{I}.
\end{aligned}$$

Summing over all Δ for a given productivity gap gives $S_{\varphi^{l},\varphi^{h}}, S_{\varphi^{l},\varphi^{l}}, S_{\varphi^{h},\varphi^{h}}, S_{\varphi^{h},\varphi^{l}}$ as stated in Proposition A-1 in the main text. It follows that

$$\begin{aligned}
\Pr\left(\Delta\leq d,\frac{\varphi^{l}}{\varphi^{h}}\right) &= \sum_{i=1}^{d}\nu\left(i,\frac{\varphi^{l}}{\varphi^{h}}\right) = S_{\varphi^{l},\varphi^{h}}\left(1-\left(\frac{I}{I+\tau}\right)^{d}\right)\\
\Pr\left(\Delta\leq d,\frac{\varphi^{l}}{\varphi^{l}}\right) &= \sum_{i=1}^{d}\nu\left(i,\frac{\varphi^{l}}{\varphi^{l}}\right) = S_{\varphi^{l},\varphi^{l}}\left(1-\left(\frac{I}{I+\tau}\right)^{d}\right)\\
\Pr\left(\Delta\leq d,\frac{\varphi^{h}}{\varphi^{h}}\right) &= \sum_{i=1}^{d}\nu\left(i,\frac{\varphi^{h}}{\varphi^{h}}\right) = S_{\varphi^{h},\varphi^{h}}\left(1-\left(\frac{I}{I+\tau}\right)^{d}\right)\\
\Pr\left(\Delta\leq d,\frac{\varphi^{h}}{\varphi^{l}}\right) &= \sum_{i=1}^{d}\nu\left(i,\frac{\varphi^{h}}{\varphi^{l}}\right) = S_{\varphi^{h},\varphi^{l}}\left(1-\left(\frac{I}{I+\tau}\right)^{d}\right).
\end{aligned}$$

Focusing on product lines where a low-productivity incumbent faces a high-productivity second-best firm:

$$P\left(\Delta\leq d,\frac{\varphi^{l}}{\varphi^{h}}\right) = S_{\varphi^{l},\varphi^{h}}\left(1-e^{-d[\ln(I+\tau)-\ln I]}\right)$$

or

$$P\left(\ln\left(\lambda^{\Delta}\right)\leq d,\frac{\varphi^{l}}{\varphi^{h}}\right) = S_{\varphi^{l},\varphi^{h}}\left(1-e^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}d}\right).$$

Conditional on the productivity gap, $\ln\left(\lambda^{\Delta}\right)$ is exponentially distributed with parameter

$\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Further

$$P\left(\lambda^{\Delta} \leq d, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - d^{-\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter $\frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. Denote $\theta = \frac{\ln(I+\tau)-\ln I}{\ln(\lambda)}$. We then have

$$P\left(\lambda^{\Delta} \leq m, \frac{\varphi^l}{\varphi^h}\right) = S_{\varphi^l,\varphi^h}\left(1 - m^{-\theta}\right).$$

Conditional on the productivity gap, quality gaps follow a Pareto distribution with parameter θ. As in Peters (2020), θ is affected by the rate of internal R&D I relative to creative destruction τ. The higher the rate of internal R&D, the more mass is in the tail of the quality gap distribution. The difference to Peters (2020) is that, in this model, quality gaps *conditional* on the productivity gap are Pareto distributed.

The following uses the obtained results to solve for the aggregate labor income share, the misallocation measure and the aggregate markup. After repeating the same steps for lines with different productivity gaps, we obtain the aggregate labor income share as follows[18]

$$\begin{aligned}\Lambda =& \sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \int_1^{\infty} \frac{1}{\varphi_k/\varphi_n} \frac{1}{m} S_{\varphi_k,\varphi_n} \theta m^{-(\theta+1)} dm \\ =& \frac{\theta}{\theta+1} \sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \frac{1}{\varphi_k/\varphi_n} S_{\varphi_k,\varphi_n}.\end{aligned} \tag{A-34}$$

The TFP misallocation statistic $\mathcal{M}$ is then given by

$$\begin{aligned}\mathcal{M} =& \frac{e^{\sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \int \left[\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}} \frac{1}{m}\right) S_{\varphi_k,\varphi_n} \theta m^{-(\theta+1)}\right] dm}}{\Lambda} \\ =& \frac{e^{\sum_{k\in\{h,l\}} \sum_{n\in\{h,l\}} \left[S_{\varphi_k,\varphi_n}\left(\ln\left(\frac{1}{\frac{\varphi_k}{\varphi_n}}\right) - \frac{1}{\theta}\right)\right]}}{\Lambda},\end{aligned}$$

where I have made use of

$$\int_1^{\infty} \ln\left(\frac{1}{m}\right) S_{\varphi_k,\varphi_n} \theta m^{-(\theta+1)} dm = \left[\frac{\theta \ln(m) + 1}{\theta m^{\theta}} + C\right]_1^{\infty} = -\frac{1}{\theta}.$$

[18] For the derivation, I assume a continuous distribution of quality gaps.

Alternatively note that this expression is equal to $-S_{\varphi_k,\varphi_n}E[\ln(\lambda^{\Delta})|\varphi_k,\varphi_n]$. I have shown above that $\ln(\lambda^{\Delta})$ conditional on the productivity gap is exponentially distributed with parameter θ. From the characteristics of an exponential distribution, its expected value is $1/\theta$.

The aggregate markup is then given by

$$\begin{aligned}E[\mu] &= \sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\int_1^{\infty}\frac{\varphi_k}{\varphi_n}mS_{\varphi_k,\varphi_n}\theta m^{-(\theta+1)}dm \\ &=\frac{\theta}{\theta-1}\sum_{k\in\{h,l\}}\sum_{n\in\{h,l\}}\frac{\varphi_k}{\varphi_n}S_{\varphi_k,\varphi_n}. \end{aligned} \tag{A-35}$$

This concludes the derivations of the aggregate labor income share, TFP misallocation statistic and aggregate markup. In Peters (2020), the step size of innovation and the rate of creative destruction relative to internal R&D, captured by θ, fully characterize the aggregate labor income share, the misallocation measure, and the aggregate markup. In this model, these statistics further depend on the size and distribution of the productivity gaps. For example, a rise in the productivity gap or a reallocation of sales shares towards high-productivity firms lowers the aggregate labor income share in eq. (A-34) and raises the aggregate markup in eq. (A-35), ceteris paribus.

D.3 Deriving the steady-state growth rate of aggregate variables

The growth rate of Q_t determines the growth rate of aggregate variables

$$g=\frac{\dot{Q}_t}{Q_t}=\frac{\partial\ln(Q_t)}{\partial t}.$$

Quality of a product in a given product line increases through internal R&D, expansion R&D or firm entry. For the growth rate of Q_t over a discrete time interval Δ, we have

$$\ln(Q_{t+\Delta})=\int_0^1\Big[(\Delta I+\Delta Sx^h+\Delta(1-S)x^l+\Delta z)\ln(\lambda)+\ln(q_{t,i})\Big]\,di$$

so that

$$\frac{\ln(Q_{t+\Delta})-\ln(Q_t)}{\Delta}=\Big(I+Sx^h+(1-S)x^l+z\Big)\ln(\lambda).$$

For $\Delta\to 0$, $g=\Big(I+Sx^h+(1-S)x^l+z\Big)\ln(\lambda)$ as stated in Proposition A-1.

D.4 Solving for the steady state equilibrium

In the model, there are the seven unknown variables $x^h, x^l, I, z, \tau, \frac{Y_t}{w_t}, S$ and the markup distribution $\nu()$ in seven equations plus the system of differential equations characterizing $\nu()$.

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} - \frac{\zeta - 1}{\psi_I} I^\zeta \right) \left(1 - \frac{1}{\lambda} \right) \frac{\psi_I}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^h)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1} \right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^l}{\varphi^h} \right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^l)^\zeta + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1} \frac{\varphi^h}{\varphi^l} \right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta \lambda^{-1} \right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}$$

Free entry condition

$$p^h\Big(SV_t^h(1, \lambda) + (1 - S)V_t^h(1, \lambda \times \varphi^h/\varphi^l)\Big) + (1 - p^h)\Big(SV_t^l(1, \lambda \times \varphi^l/\varphi^h) + (1 - S)V_t^l(1, \lambda)\Big) = \frac{1}{\psi_z} w_t,$$

where

$$V_t^d(1, \mu) = \frac{1}{(\rho + \tau)}\frac{\zeta - 1}{\psi_x}(x^d)^\zeta w_t + \frac{Y_t\left(1 - \frac{1}{\mu}\right) + \frac{\zeta - 1}{\psi_I} I^\zeta w_t \mu^{-1}}{\rho + \tau}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t} \sum_{\frac{\varphi_j}{\varphi_{j'}}} \sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}} \nu\left(i, \frac{\varphi_j}{\varphi_{j'}}\right) + \frac{1}{\psi_I} I^\zeta \sum_{\frac{\varphi_j}{\varphi_{j'}}} \sum_i \frac{1}{\lambda^i \frac{\varphi_j}{\varphi_{j'}}} \nu\left(i, \frac{\varphi_j}{\varphi_{j'}}\right) + S\frac{1}{\psi_x}(x^h)^\zeta + (1 - S)\frac{1}{\psi_x}(x^l)^\zeta + \frac{z}{\psi_z}$$

Creative destruction

$$\tau = z + Sx^h + (1 - S)x^l$$

Share of high productivity type

$$S = \sum_{i=1}^{\infty} \left[\nu \left(i, \frac{\varphi^h}{\varphi^h} \right) + \nu \left(i, \frac{\varphi^h}{\varphi^l} \right) \right],$$

where ν, the stationary distribution of quality and productivity gaps, is characterized by

$$0 = \dot{\nu} \left(\Delta, \frac{\varphi_j}{\varphi_{j'}} \right) = I\nu \left(\Delta - 1, \frac{\varphi_j}{\varphi_{j'}} \right) - \nu \left(\Delta, \frac{\varphi_j}{\varphi_{j'}} \right) (I + \tau) \quad \text{for} \quad \Delta \geq 2$$

and for the case of a unitary quality gap

$$\begin{aligned}
0 = \dot{\nu} \left(1, \frac{\varphi^l}{\varphi^h} \right) &= (1 - S)x^l S + z_t(1 - p^h)S - \nu \left(1, \frac{\varphi^l}{\varphi^h} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^l}{\varphi^l} \right) &= (1 - S)x^l(1 - S) + z_t(1 - p^h)(1 - S) - \nu \left(1, \frac{\varphi^l}{\varphi^l} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^h}{\varphi^h} \right) &= Sx^h S + z_t p^h S - \nu \left(1, \frac{\varphi^h}{\varphi^h} \right) (I + \tau) \\
0 = \dot{\nu} \left(1, \frac{\varphi^h}{\varphi^l} \right) &= Sx^h(1 - S) + z_t p^h(1 - S) - \nu \left(1, \frac{\varphi^h}{\varphi^l} \right) (I + \tau).
\end{aligned}$$

To simplify the system of equations, first rewrite the rate of creative destruction

$$z = (\tau - Sx^h - (1 - S)x^l)$$

such that z can be substituted out from the remaining equations. Second, based on Proposition A-1, we know

$$S = \frac{Sx^h + zp^h}{\tau}.$$

Third, the optimality conditions for expansion rates (multiplied by p^h and $(1 - p^h)$) and the free entry condition together imply

$$\frac{1}{\psi_x} p^h (x^h)^{\zeta - 1} + \frac{1}{\psi_x} (1 - p^h)(x^l)^{\zeta - 1} = \frac{1}{\psi_z \zeta}.$$

The system of equilibrium conditions can hence be reduced to:

Optimality condition for the internal innovation rate

$$I = \left(\left(\frac{Y_t}{w_t} \psi_I - (\zeta - 1) I^{\zeta} \right) \frac{\left(1 - \frac{1}{\lambda}\right)}{\zeta(\rho + \tau)} \right)^{\frac{1}{\zeta - 1}}$$

Optimality condition for high-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^h)^{\zeta} + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{1}{\psi_I}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{\varphi^l}{\varphi^h}\frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{\varphi^l}{\psi_I \varphi^h}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^h)^{\zeta - 1}$$

Optimality condition for low-productivity type expansion rate

$$\frac{\zeta - 1}{\psi_x}(x^l)^{\zeta} + S\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\frac{\varphi^h}{\varphi^l}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{\varphi^h}{\psi_I \varphi^l}\right) + (1 - S)\left(\frac{Y_t}{w_t}\left(1 - \frac{1}{\lambda}\right) + (\zeta - 1)I^{\zeta}\lambda^{-1}\frac{1}{\psi_I}\right)$$
$$= (\rho + \tau)\frac{\zeta}{\psi_x}(x^l)^{\zeta - 1}$$

Free entry

$$p^h \frac{(x^h)^{\zeta - 1}}{\psi_x} + (1 - p^h)\frac{(x^l)^{\zeta - 1}}{\psi_x} = \frac{1}{\psi_z \zeta}$$

Labor market clearing condition

$$1 = \frac{Y_t}{w_t}\Lambda + \Lambda_I + S\frac{1}{\psi_x}(x^h)^{\zeta} + (1 - S)\frac{1}{\psi_x}(x^l)^{\zeta} + \frac{\tau - Sx^h - (1 - S)x^l}{\psi_z},$$

where

$$\begin{aligned}
\Lambda =& \frac{\theta}{\theta + 1} \sum_{k \in \{h,l\}} \sum_{n \in \{h,l\}} \frac{1}{\varphi_k / \varphi_n} S_{\varphi_k, \varphi_n} \\
\Lambda_I =& \frac{1}{\psi_I} I^{\zeta} \frac{\theta}{\theta + 1} \sum_{k \in \{h,l\}} \sum_{n \in \{h,l\}} \frac{1}{\varphi_k / \varphi_n} S_{\varphi_k, \varphi_n} \\
\theta =& \frac{\ln(I + \tau) - \ln(I)}{\ln(\lambda)}
\end{aligned}$$

Sales share of high productivity type

$$S = \frac{Sx^h + (\tau - Sx^h - (1 - S)x^l)p^h}{\tau}$$

The expressions related to the labor market clearing condition are derived in Section D.2. This constitutes a system of seven equations in seven unknowns $(x^h, x^l, I, \tau, \frac{Y_t}{w_t}, S)$, which I solve using a root finder.

D.5 Firm markups

Firm markups are defined by $\mu_f = \frac{py_f}{wl_f} = \left(\frac{1}{n}\sum_{k=1}^{n}\mu_{kf}^{-1}\right)^{-1}$. Therefore

$$\ln \mu_f = -\ln\left(\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1}\right).$$

Rewrite the term in brackets (for a high-productivity firm) as

$$\frac{1}{n}\sum_{k=1}^{n}\mu_k^{-1} = \frac{1}{n}\sum_{k=1}^{n}e^{-\ln\mu_k} = \frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right), \tag{A-36}$$

where i indexes the product lines where the high productivity firm faces a low productivity second best producer, j the lines where it faces a high productivity second best producer and $n_i + n_j = n$. A two-dimensional linear Taylor expansion around $\ln\lambda = 0$ and $\ln\frac{\varphi^h}{\varphi^l} = 0$ gives

$$\frac{1}{n}\left(\sum_{i=1}^{n_i}e^{-\ln\frac{\varphi^h}{\varphi^l}-\Delta_i\ln\lambda} + \sum_{j=1}^{n_j}e^{-\Delta_j\ln\lambda}\right) \approx 1 - \left(\frac{1}{n}\sum_{k=1}^{n}\Delta_k\right)\ln\lambda - \frac{n_i}{n}\ln\left(\frac{\varphi^h}{\varphi^l}\right)$$

such that

$$E\left[\ln\mu_f|\text{firm age} = a_f, \varphi^h\right] \approx E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda + (1-S)\ln\left(\frac{\varphi^h}{\varphi^l}\right),$$

where I have used the fact that (in expectation) the share of the firm's product lines with a low productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the low productivity type. From Peters (2020), we know that

$$E\left[\frac{1}{n}\sum_{k=1}^{n}\Delta_k|\text{firm age} = a_f, \varphi^h\right]\ln\lambda = \left(1 + I \times E[a_P^h|a_f]\right)\ln\lambda,$$

where $E[a_P^h|a_f]$ denotes the average product age of a high-productivity type firm conditional on firm age a_f and

$$E[a_P^h|a_f] = \frac{1}{x^h}\left(\frac{\frac{1}{\tau}\left(1-e^{-\tau a_f}\right)}{\frac{1}{x^h+\tau}\left(1-e^{-(x^h+\tau)a_f}\right)} - 1\right)\left(1-\phi^h(a_f)\right) + a_f\phi^h(a_f)$$

$$\phi^h(a) = e^{-x^h a} \frac{1}{\gamma^h(a)} \ln \left(\frac{1}{1-\gamma^h(a)} \right)$$
$$\gamma^h(a) = \frac{x^h \left(1 - e^{-(\tau - x^h)a}\right)}{\tau - x^h e^{-(\tau-x^h)a}}.$$

For a firm of the low-productivity type, the last term in eq. (A-36) reads

$$\frac{1}{n} \left(\sum_{i=1}^{n_i} e^{-\ln \Delta_i \ln \lambda} + \sum_{j=1}^{n_j} e^{-\ln \frac{\varphi^l}{\varphi^h} - \ln \Delta_j \ln \lambda} \right),$$

where i indexes the product lines where the low-productivity producer faces a low-productivity second best producer, j the lines where it faces a high-productivity second best producer and $n_i + n_j = n$. Following the same steps as for a high-productivity firm, this time linearizing around $\ln \frac{\varphi^l}{\varphi^h} = 0$ (and $\ln \lambda = 0$) gives

$$E\left[\ln \mu_f | \text{firm age} = a_f, \varphi^l\right] \approx \left(1 + I \times E[a_P^l | a_f]\right) \ln \lambda + S \ln \left(\frac{\varphi^l}{\varphi^h}\right),$$

where again I have made use of the fact that (in expectation) the share of the firm's product lines with a high-productivity second best producer is equal to the aggregate share of product lines where the incumbent is of the high-productivity type. $E[a_P^l|a_f]$ is exactly defined as $E[a_P^h|a_f]$ with x^h replaced by x^l in the above expressions.

D.6 Markup-age relationship

As show in the Appendix, Section D.5 the expected log markup conditional on firm age a_f for a high-productivity type firm is

$$E\left[\ln \mu_f | a_f, \varphi^h\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^h|a_f]\right)}_{\text{Quality improvements}} + \underbrace{(1-S) \times \ln \left(\frac{\varphi^h}{\varphi^l}\right)}_{\text{Productivity advantage}}, \tag{A-37}$$

where $E[a_P^h|a_f]$ is the average product age of a high-type firm conditional on firm age.

The expected firm markup conditional on age consists of two terms. The first term in eq. (A-37) is akin to Peters (2020) and reflects that internal R&D at the product level translates into markup growth at the firm level as it ages. In Peters (2020), this term holds for all firms, whereas in this model, this term is specific to the productivity type of the firm, as the average product age varies by firm type. The second term in eq. (A-37) captures a new level effect that heterogeneity in productivity introduces. The intuition is that if a high-type incumbent faces a low-type second-best firm, it can charge a φ^h/φ^l higher markup, which occurs in expectation in $1-S$ of the incumbent's product lines. The markup-age relationship for high-productivity firms then equals $E\left[\ln \mu_f | a_f, \varphi^h\right] - E\left[\ln \mu_f | 0, \varphi^h\right]$.

The expected markup conditional on firm age for a low-productivity type firm follows

$$E\left[\ln \mu_f | a_f, \varphi^l\right] = \underbrace{\ln \lambda \times \left(1 + I \times E[a_P^l | a_f]\right)}_{\text{Quality improvements}} + \underbrace{S \times \ln\left(\frac{\varphi^l}{\varphi^h}\right)}_{\text{Productivity disadvantage}}. \tag{A-38}$$

The first term captures quality improvements through internal R&D. The second term in eq. (A-38) differs from eq. (A-37). Low-productivity incumbents face a high-productivity second-best firm in a share S of their product lines. Since $\varphi^l < \varphi^h$, this term is negative. The markup-age relationship is then defined likewise as for high-productivity firms.

D.7 Firm size distribution

The mass of high- and low-productivity type firms with $n \geq 2$ products follows the differential equations

$$\begin{aligned} \dot{M}_t^h(n) &= (n-1)x_t^h M_t^h(n-1) + (n+1)\tau_t M_t^h(n+1) - n(x_t^h + \tau_t)M_t^h(n) \\ \dot{M}_t^l(n) &= (n-1)x_t^l M_t^l(n-1) + (n+1)\tau_t M_t^l(n+1) - n(x_t^l + \tau_t)M_t^l(n), \end{aligned} \tag{A-39}$$

whereas the mass of firms with one product evolves according to

$$\begin{aligned} \dot{M}_t^h(1) &= z_t p^h + 2\tau_t M_t^h(2) - (x_t^h + \tau_t)M_t^h(1) \\ \dot{M}_t^l(1) &= z_t(1-p^h) + 2\tau_t M_t^l(2) - (x_t^l + \tau_t)M_t^l(1). \end{aligned} \tag{A-40}$$

The mass of firms with n products increases through firms with $n-1$ products expanding to size n at rate x_t^h or x_t^l per product or through firms with $n+1$ products losing a product at the rate of aggregate creative destruction τ_t. The mass of firms with n products decreases through firms with n products either gaining or losing a product through expansion or creative destruction. The mass of firms with one product additionally increases through firm entry.

Proposition A-2. *The stationary firm size distribution along the balanced growth path is characterized as follows.*

1. *The mass of high and low productivity firms with n products is*

$$M^h(n) = \frac{(x^h)^{n-1} z p^h}{n\tau^n} = \frac{zp^h}{x^h}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n$$
$$M^l(n) = \frac{(x^l)^{n-1} z(1-p^h)}{n\tau^n} = \frac{z(1-p^h)}{x^l}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n.$$

2. *The total mass of firms with n products is*

$$M(n) = M^h(n) + M^l(n) = \frac{(x^h)^{n-1} z p^h + (x^l)^{n-1} z(1-p^h)}{n\tau^n}.$$

3. *The mass of firms of each productivity type is*

$$M^h = \sum_{n=1}^{\infty} M^h(n) = \frac{zp^h}{x^h}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^h}{\tau}\right)^n = \frac{zp^h}{x^h}\ln\left(\frac{\tau}{\tau - x^h}\right)$$
$$M^l = \sum_{n=1}^{\infty} M^l(n) = \frac{z(1-p^h)}{x^l}\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{x^l}{\tau}\right)^n = \frac{z(1-p^h)}{x^l}\ln\left(\frac{\tau}{\tau - x^l}\right)$$

4. *The total mass of firms is*

$$M = M^h + M^l.$$

Proof. These results follow from setting the time derivatives in equations (A-39) and (A-40) equal to zero and solving the system of equations. □

For each firm type, the share of firms with n products, $M^h(n)/M^h$ and $M^l(n)/M^l$, follows the PDF of a logarithmic distribution with parameter x^h/τ and x^l/τ as in Lentz and Mortensen (2008). The firm size distribution is highly skewed to the right.

Since there is a continuum of mass one of products and each product is mapped to one firm $\sum_{i=1}^{\infty} M(n) \times n = 1$. Further, the mass of high-productivity type firms producing n products is related to the share of product lines operated by high-type firms, S, as follows

$$S = \sum_{n=1}^{\infty} M^h(n) \times n = \frac{zp^h}{\tau - x^h}.$$